Supplementary Material for "Randomization Inference beyond the Sharp Null: Bounded Null Hypotheses and Quantiles of Individual Treatment Effects"

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A1. Further discussion of primary theoretical results

A1.1. Extended discussion of the tie-breaking methods

We first discuss the subtle issue of defining ranks when there exist ties. There are multiple ways to define ranks for ties, see, e.g., the R documentation for the function rank (R Core Team 2013). Consider an arbitrary outcome vector $\boldsymbol{y} \in \mathbb{R}^n$. The method "random" puts equal values of y_i's in a random order, the methods "first" and "last" rank equal values of y_i's based on their indices in an increasing or decreasing way[1], and the method "average" replaces their ranks by the corresponding average. The first three methods will always produce ranks from 1 to n, which is important for the distribution free property in Definition 4, while the ranks from the last generally depend on the value of $\boldsymbol{y}$.

We then give a more detailed discussion of Proposition 2(b). The statistic $t_2(\cdot, \cdot)$ in (6) is distribution free under the "random" method for ties. In practice, we can implement the "random" method by first randomly permuting the coordinates of $\boldsymbol{y}$ and then using the "first" or "last" method. We emphasize that using the "random" method for ties is crucial. First, a defined method for ties is important even for continuous outcomes, because some outcomes will become the same when we later invert tests for, e.g., a sequence of constant treatment effects. Second, it is important to rank equal outcomes randomly, since in practice a received data set may be ordered, e.g., with all treated units ahead of control ones.

We finally give some discussion regarding Proposition 2(a). For the usual "average" method for ties, whether the statistic in (6) is effect increasing depends on the values of $\phi(\cdot)$ for tied ranks. If we define the value of $\phi(\cdot)$ for tied ranks as the average value of $\phi(\cdot)$ evaluated at those ranks under the "first" or "last" method, then the statistic (6) is effect increasing. This follows directly from Proposition 2 by noting that the resulting value of the statistic is essentially the average of it under the "first" method of ties over all possible permutations of the ordering. From the above, we can also know that the classical Wilcoxon rank sum statistic with "average" method for ties is effect increasing.

A1.2. Discussion of Theorem 1 in Section 4.1

Theorem 1(b) presents a stronger conclusion than (a), but with stronger condition on the test statistic. Specifically, for any $\boldsymbol{\delta} \in \mathbb{R}^n$, if the bounded null $H_{\preccurlyeq \boldsymbol{\delta}}$ holds (i.e., $\boldsymbol{\tau} \preccurlyeq \boldsymbol{\delta}$), then Theorem 1(b) implies that the randomization p-value $p_{\boldsymbol{Z},\boldsymbol{\delta}} \geq p_{\boldsymbol{Z},\boldsymbol{\tau}}$ and is thus stochastically larger than or equal to Unif$[0, 1]$. More importantly, Theorem 1(b) is useful for constructing confidence sets for the true treatment effect $\boldsymbol{\tau}$, as discussed in the main paper. In principle, one could invert

[1] For example, if we rank the coordinates of $\boldsymbol{y}$ using the "first" method, then $\mathrm{r}_i(\boldsymbol{y}) < \mathrm{r}_j(\boldsymbol{y})$ if and only if (a) $y_i < y_j$ or (b) $y_i = y_j$ and $i < j$.

randomization tests for all possible sharp null hypotheses $H_{\boldsymbol{\delta}}$'s to get confidence sets for the true treatment effect $\boldsymbol{\tau}$. However, enumerating all possible sharp null hypotheses is generally computationally intractable, except in the cases of binary or discrete outcomes (see Rigdon and Hudgens 2015). For discrete outcomes where enumeration is possible, Theorem 1(b) can help reduce the number of enumerations (Li and Ding 2016).

A1.3. Discussion of Theorems 4 and 5 and Corollary 2 in Sections 5.3–5.5

Theorem 4 and Corollary 2 provide confidence intervals for quantiles of individual effects $\tau_{(k)}$'s as well as number (or equivalently proportion) of units with effects larger than any threshold. However, both theorems do not provide point estimation for these quantities. Indeed, these quantities are generally not identifiable due to no joint observation of the treatment and control potential outcomes for any unit. Consequently, consistent estimators for them generally do not exist. Recently, for binary or ordinal outcomes, Lu et al. (2018) and Huang et al. (2019) studied sharp bounds and constructed confidence intervals for the proportions of units with positive effects, i.e, $n(0)/n$. Importantly, our confidence intervals in Corollary 2 work for general outcomes.

Furthermore, we could, in principle, construct confidence sets for the n-dimensional individual effect vector $\boldsymbol{\tau}$ by inverting tests for all sharp null hypotheses. In general, however, testing arbitrary sharp null hypotheses is computationally infeasible, and it does not provide informative inferences because the parameter space is typically too unwieldy (with n units, the space of possible effects is n-dimensional). As noted by Rosenbaum (2010), such a confidence set would not be intelligible, since it would be a subset of an n-dimensional space. Some special forms of outcomes or test statistics can lead to efficient computation and intuitive confidence sets. For example, Rosenbaum (2001) used carefully designed test statistics involving the attributable effects, and Rigdon and Hudgens (2015) considered binary outcomes. A key property utilized by these approaches is that many sharp null hypotheses are equally likely in the sense of producing the same randomization p-value, which not only avoids enumeration over all possible values of $\boldsymbol{\tau}$ but also provides a convenient form for the resulting confidence sets. By contrast, our confidence sets constructed in Theorem 5 work for general outcomes and rely mainly on the valid p-value (13) for testing null hypotheses about quantiles of individual effects, which involves efficient optimization of randomization p-value in Theorem 3. Moreover, as illustrated in Section 1.2, Theorem 5 provides nice confidence sets in $\mathbb{R}^n$ that are easy to understand, interpret and visualize.

In fact, our confidence set in Theorem 5 essentially provides a confidence band for the quantile (or equivalently distribution) function of the individual treatment effects. Relatedly, Lei and Candès (2021) constructed prediction intervals for the (random) individual treatment effect, assuming random sampling of units from some superpopulation. One main difference is that we focus on the fixed population distribution of individual effects, while they focus on a random draw from the superpopulation distribution of individual effects.

A2. Broader justification for Imbens and Rubin (2015) style Fisher randomization tests

First, as mentioned in the main paper, it should not be surprising that we can also use test statistics of the form $t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(1))$, which involves the imputed treatment potential outcomes instead of imputed control ones. This can be achieved by switching the labels of treatment and control and changing the signs of the outcomes.

More broadly, the Fisher randomization tests in Imbens and Rubin (2015) use test statistics of form $t(\boldsymbol{Z}, \boldsymbol{Y})$, which often compare the observed outcomes of treated units to those of control units (e.g., the difference in outcome means between treatment and control groups). Analogous to the main paper, such a randomization test can also be valid for testing bounded null hypotheses as we discuss below. Unfortunately, our generalization for inferring quantiles of individual effects (as in Section 5) relies crucially on the use of test statistics of form $t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))$; our results do not extend to this class of statistic. This is why we focus on the randomization p-value of form (4) in the main paper.

That being said, we next provide the broader justification for the Fisher randomization test in Imbens and Rubin (2015), which differs from that in Rosenbaum (2002) in the choice of test statistic. We first formally describe the randomization p-value in Imbens and Rubin (2015), and then study its property in parallel with that in Section 4.

A2.1. The randomization p-value in Imbens and Rubin (2015)

The Fisher randomization test in Imbens and Rubin (2015) considers test statistics of form $t(\boldsymbol{Z}, \boldsymbol{Y})$, which often compare the observed outcomes of treated units to those of control units. For the null $H_{\boldsymbol{\delta}}$ in (1), the imputed potential outcomes are $\boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(1)$ and $\boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)$ in (2). For any treatment assignment vector $\boldsymbol{a} \in \mathcal{Z}$, the corresponding imputed observed outcome vector would then be

$$\boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(\boldsymbol{a}) \equiv \boldsymbol{a} \circ \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(1) + (1 - \boldsymbol{a}) \circ \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0),$$

and the corresponding test statistic would have value $t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(\boldsymbol{a}))$. Thus, for the null $H_{\boldsymbol{\delta}}$, the imputed randomization distribution of the test statistic has the following tail probability:

$$\tilde{G}_{\boldsymbol{Z},\boldsymbol{\delta}}(c) \equiv \Pr\left\{t(\boldsymbol{A}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(\boldsymbol{A})) \geq c\right\} = \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(\boldsymbol{a})) \geq c\right\}, \qquad (c \in \mathbb{R}) \tag{A1}$$

and the corresponding randomization p-value is the tail probability (A1) evaluated at the observed value of the test statistic:

$$\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}} \equiv \tilde{G}_{\boldsymbol{Z},\boldsymbol{\delta}}(t(\boldsymbol{Z}, \boldsymbol{Y})) = \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(\boldsymbol{a})) \geq t(\boldsymbol{Z}, \boldsymbol{Y})\right\}. \tag{A2}$$

When $H_{\boldsymbol{\delta}}$ is true, the imputed randomization distribution $G_{\boldsymbol{Z},\boldsymbol{\delta}}(\cdot)$ in (A1) is the same as the true one, and the randomization p-value $\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}}$ in (A2) is stochastically larger than or equal to Unif$[0, 1]$, i.e., it is a valid p-value for testing $H_{\boldsymbol{\delta}}$.

The two p-values $\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}}$ in (A2) and $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ in (4) are equivalent for testing the Fisher's null $H_{\mathbf{0}}$ of no effect. As commented in Appendix A4.2, they are also equivalent for testing the null $H_{c\mathbf{1}}$ of constant effect c for any $c \in \mathbb{R}$, if both of them use difference-in-means as the test statistic. However, they are generally not equivalent. Two obvious differences are as follows. First, given the observed data $(\boldsymbol{Z}, \boldsymbol{Y})$, $\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}}$ depends on the speculation of individual effects for all units, while $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ depends only on the speculation of individual effects for treated units or equivalently $\boldsymbol{Z} \circ \boldsymbol{\delta}$. Second, for $\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}}$, the tail probability $\tilde{G}_{\boldsymbol{Z},\boldsymbol{\delta}}(\cdot)$ in (A1) depends on the null of interest $\boldsymbol{\delta}$, but the cutoff $t(\boldsymbol{Z}, \boldsymbol{Y})$ does not depend on $\boldsymbol{\delta}$; while for $p_{\boldsymbol{Z},\boldsymbol{\delta}}$, both the tail probability $G_{\boldsymbol{Z},\boldsymbol{\delta}}(\cdot)$ in (3) and the cutoff $t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))$ depend on $\boldsymbol{\delta}$.

A2.2. Validity of the randomization p-value for testing bounded nulls

Below we study the randomization $\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}}$ in (A2) for testing bounded nulls, its monotonicity property and its usage for constructing confidence intervals for the maximum individual effect. We summarize the results in the following theorem and corollary, in parallel with Theorem 1 and Corollary 1 in the main paper for the randomization p-value $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ in (4).

Theorem A1. If the statistic $t(\cdot, \cdot)$ is effect increasing, then

(a) for any constant $\boldsymbol{\delta} \in \mathbb{R}^n$, the corresponding randomization p-value $\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}}$ in (A2) for the sharp null $H_{\boldsymbol{\delta}}$ in (1) is also valid for testing the bounded null $H_{\preccurlyeq\boldsymbol{\delta}}$ in (7). Specifically, under $H_{\preccurlyeq\boldsymbol{\delta}}$, $\Pr(\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}} \le \alpha) \le \alpha$ for any $\alpha \in (0, 1)$;

(b) for any possible assignment $\boldsymbol{z} \in \mathcal{Z}$, the corresponding randomization p-value $\tilde{p}_{\boldsymbol{z},\boldsymbol{\delta}}$ in (A2), viewed as a function of $\boldsymbol{\delta} \in \mathbb{R}^n$, is monotone increasing. Specifically, $\tilde{p}_{\boldsymbol{z},\boldsymbol{\delta}} \le \tilde{p}_{\boldsymbol{z},\overline{\boldsymbol{\delta}}}$ for any $\boldsymbol{\delta} \preccurlyeq \overline{\boldsymbol{\delta}}$.

Corollary A1. (a) If the test statistic $t(\cdot, \cdot)$ is effect increasing, then for any $\alpha \in (0, 1)$, the set $\{c : \tilde{p}_{\boldsymbol{Z},c\mathbf{1}} > \alpha, c \in \mathbb{R}\}$ is a $1 - \alpha$ confidence set for the maximum individual effect $\tau_{\max}$. (b) Furthermore, the confidence set must have the form of $(\underline{c}, \infty)$ or $[\underline{c}, \infty)$ with $\underline{c} = \inf\{c : \tilde{p}_{\boldsymbol{Z},c\mathbf{1}} > \alpha, c \in \mathbb{R}\}$.

A3. Proofs for properties of test statistics

Proof of Proposition 1. Suppose that both $\psi_{1i}(\cdot)$ and $\psi_{0i}(\cdot)$ are monotone increasing functions for all $1 \le i \le n$. We first show that the statistic $t_1(\cdot, \cdot)$ in (5) is effect increasing. By definition, for any $\boldsymbol{z} \in \mathcal{Z}$ and $\boldsymbol{y}, \boldsymbol{\eta}, \boldsymbol{\xi} \in \mathbb{R}^n$ with $\boldsymbol{\eta} \succcurlyeq \mathbf{0} \succcurlyeq \boldsymbol{\xi}$,

$$t_1(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{z} \circ \boldsymbol{\eta} + (\mathbf{1} - \boldsymbol{z}) \circ \boldsymbol{\xi}) - t_1(\boldsymbol{z}, \boldsymbol{y})$$

$$
\begin{aligned}
&= \left\{ \sum_{i=1}^{n} z_i \psi_{1i}(y_i + \eta_i) - \sum_{i=1}^{n} (1 - z_i) \psi_{0i}(y_i + \xi_i) \right\} - \left\{ \sum_{i=1}^{n} z_i \psi_{1i}(y_i) - \sum_{i=1}^{n} (1 - z_i) \psi_{0i}(y_i) \right\} \\
&= \sum_{i=1}^{n} z_i \left\{ \psi_{1i}(y_i + \eta_i) - \psi_{1i}(y_i) \right\} + \sum_{i=1}^{n} (1 - z_i) \left\{ \psi_{0i}(y_i) - \psi_{0i}(y_i + \xi_i) \right\}. \qquad \text{(A3)}
\end{aligned}
$$

Because $\eta_i \geq 0 \geq \xi_i$ and $\psi_{1i}(\cdot)$ and $\psi_{1i}(\cdot)$ are monotone increasing functions for all i, we can know that all the terms in (A3) are non-negative. Therefore, (A3) must also be non-negative. From Definition 2, $t_1(\cdot, \cdot)$ in (5) is effect increasing.

We then show that the statistic $t_1(\cdot, \cdot)$ in (5) is differential increasing. By definition, for any $\boldsymbol{z}, \boldsymbol{a} \in \mathcal{Z}$ and $\boldsymbol{y}, \boldsymbol{\eta} \in \mathbb{R}^n$ with $\boldsymbol{\eta} \succcurlyeq 0$,

$$
\begin{aligned}
& t_1(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - t_1(\boldsymbol{z}, \boldsymbol{y}) \\
&= \left\{ \sum_{i=1}^{n} z_i \psi_{1i}(y_i + a_i \eta_i) - \sum_{i=1}^{n} (1 - z_i) \psi_{0i}(y_i + a_i \eta_i) \right\} - \left\{ \sum_{i=1}^{n} z_i \psi_{1i}(y_i) - \sum_{i=1}^{n} (1 - z_i) \psi_{0i}(y_i) \right\} \\
&= \sum_{i=1}^{n} z_i \left\{ \psi_{1i}(y_i + a_i \eta_i) - \psi_{1i}(y_i) + \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\} - \sum_{i=1}^{n} \left\{ \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\}. \quad \text{(A4)}
\end{aligned}
$$

For each $1 \leq i \leq n$, because $a_i \in \{0, 1\}$, $\eta_i \geq 0$, and both $\psi_{1i}(\cdot)$ and $\psi_{1i}(\cdot)$ are increasing functions, we can know that

$$
\begin{aligned}
& z_i \left\{ \psi_{1i}(y_i + a_i \eta_i) - \psi_{1i}(y_i) + \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\} \\
& \begin{cases} \leq a_i \left\{ \psi_{1i}(y_i + a_i \eta_i) - \psi_{1i}(y_i) + \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\}, & \text{if } a_i = 1, \\ = 0 = a_i \left\{ \psi_{1i}(y_i + a_i \eta_i) - \psi_{1i}(y_i) + \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\}, & \text{if } a_i = 0. \end{cases}
\end{aligned}
$$

This immediately implies that (A4) can be bounded above by

$$
\begin{aligned}
& t_1(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - t_1(\boldsymbol{z}, \boldsymbol{y}) \\
&= \sum_{i=1}^{n} z_i \left\{ \psi_{1i}(y_i + a_i \eta_i) - \psi_{1i}(y_i) + \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\} - \sum_{i=1}^{n} \left\{ \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\} \\
&\leq \sum_{i=1}^{n} a_i \left\{ \psi_{1i}(y_i + a_i \eta_i) - \psi_{1i}(y_i) + \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\} - \sum_{i=1}^{n} \left\{ \psi_{0i}(y_i + a_i \eta_i) - \psi_{0i}(y_i) \right\} \\
&= t_1(\boldsymbol{a}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - t_1(\boldsymbol{a}, \boldsymbol{y}).
\end{aligned}
$$

From Definition 3, $t_1(\cdot, \cdot)$ in (5) is differential increasing.

From the above, Proposition 1 holds. □

To prove Proposition 2, we need the following three lemmas.

Lemma A1. For any $n \geq 1$, $\boldsymbol{y} \in \mathbb{R}^n$, for any $1 \leq i, j \leq n$, define

$$\delta_{ij}(y_i, y_j) = \begin{cases} \mathbb{1}(y_i > y_j) + \mathbb{1}(y_i = y_j)\mathbb{1}(i \geq j), & \text{if "first" method is used for ties,} \\ \mathbb{1}(y_i > y_j) + \mathbb{1}(y_i = y_j)\mathbb{1}(i \leq j), & \text{if "last" method is used for ties.} \end{cases} \tag{A5}$$

Then for $1 \leq i \leq n$, $\mathrm{r}_i(\boldsymbol{y}) = \sum_{j=1}^{n} \delta_{ij}(y_i, y_j)$.

Proof of Lemma A1. Lemma A1 follows directly from the definition of ranks. □

Lemma A2. $\delta_{ij}(\cdot, \cdot)$'s defined in (A5) have the following properties:

(a) for any (i, j), $\delta_{ij}(x, y)$ is increasing in x and is decreasing in y;

(b) for any (i, j, l), if $\delta_{ij}(x, y) = 1$ and $\delta_{jl}(y, z) = 1$, then we must have $\delta_{il}(x, z) = 1$.

Proof of Lemma A2. Lemma A2 follows directly from the definition of $\delta_{ij}(\cdot, \cdot)$'s. □

Lemma A3. For any $n \geq 1$, $\boldsymbol{z} \in \{0, 1\}^n$, and $\boldsymbol{y}, \boldsymbol{\eta}, \boldsymbol{\xi} \in \mathbb{R}^n$ with $\boldsymbol{\eta} \succcurlyeq \mathbf{0} \succcurlyeq \boldsymbol{\xi}$, if $\phi(\cdot)$ is a monotone increasing function and "first" or "last" method is used for ties, then

$$\sum_{i=1}^{n} z_i \phi\left(\mathrm{r}_i(\boldsymbol{y} + \boldsymbol{z} \circ \boldsymbol{\eta} + (\mathbf{1} - \boldsymbol{z}) \circ \boldsymbol{\xi})\right) \geq \sum_{i=1}^{n} z_i \phi(\mathrm{r}_i(\boldsymbol{y})).$$

Proof of Lemma A3. Let $m = \sum_{i=1}^{n} z_i$, and $\boldsymbol{w} = \boldsymbol{y} + \boldsymbol{z} \circ \boldsymbol{\eta} + (\mathbf{1} - \boldsymbol{z}) \circ \boldsymbol{\xi}$. We use $(i_1, i_2, \ldots, i_m)$ to denote permutation of the set $\{i : z_i = 1, 1 \leq i \leq n\}$ such that $\mathrm{r}_{i_1}(\boldsymbol{y}) < \mathrm{r}_{i_2}(\boldsymbol{y}) < \ldots < \mathrm{r}_{i_m}(\boldsymbol{y})$, and $(j_1, j_2, \ldots, j_m)$ to denote permutation of the set $\{i : z_i = 1, 1 \leq i \leq n\}$ such that $\mathrm{r}_{j_1}(\boldsymbol{w}) < \mathrm{r}_{j_2}(\boldsymbol{w}) < \ldots < \mathrm{r}_{j_m}(\boldsymbol{w})$. For each $1 \leq k \leq m$, from Lemma A1, we have

$$\mathrm{r}_{i_k}(\boldsymbol{y}) = \sum_{l=1}^{n} \delta_{i_k l}(y_{i_k}, y_l) = \sum_{l: z_l = 1} \delta_{i_k l}(y_{i_k}, y_l) + \sum_{l: z_l = 0} \delta_{i_k l}(y_{i_k}, y_l) = k + \sum_{l: z_l = 0} \delta_{i_k l}(y_{i_k}, y_l), \tag{A6}$$

where the last equality holds due to the construction of i_k. By the same logic, for $1 \leq k \leq m$,

$$\mathrm{r}_{j_k}(\boldsymbol{w}) = k + \sum_{l: z_l = 0} \delta_{j_k l}(w_{j_k}, w_l). \tag{A7}$$

First, we prove $\delta_{j_k i_k}(w_{j_k}, y_{i_k}) = 1$ for all $1 \leq k \leq m$. For $1 \leq p \leq k$, because $w_{j_p} = y_{j_p} + \eta_{j_p} \geq y_{j_p}$, from Lemma A2(a), we have $\delta_{j_k j_p}(w_{j_k}, y_{j_p}) \geq \delta_{j_k j_p}(w_{j_k}, w_{j_p}) = 1$. By the definition of i_k, there must exist a $j_q \in \{j_1, j_2, \ldots, j_k\}$ such that $\delta_{j_q i_k}(y_{j_q}, y_{i_k}) = 1$. Thus, by Lemma A2(b), we must have $\delta_{j_k i_k}(w_{j_k}, y_{ik}) = 1$.

Second, we prove that $\mathrm{r}_{j_k}(\boldsymbol{w}) \geq \mathrm{r}_{i_k}(\boldsymbol{y})$ for all $1 \leq k \leq m$. By definition, for any l with $z_l = 0$, we have $w_l = y_l + \xi_l \leq y_l$. Thus, Lemma A2(a) implies that $\delta_{j_k l}(w_{j_k}, w_l) \geq \delta_{j_k l}(w_{j_k}, y_l)$ for any l with $z_l = 0$. From the previous discussion, $\delta_{j_k i_k}(w_{j_k}, y_{i_k}) = 1$ for all $1 \leq k \leq m$. From Lemma A2(b), for any $1 \leq k \leq m$ and l with $z_l = 0$, if $\delta_{i_k l}(y_{i_k}, y_l) = 1$, then we must have

$\delta_{j_k l}(w_{j_k}, w_l) \geq \delta_{j_k l}(w_{j_k}, y_l) = 1$. Consequently, $\delta_{j_k l}(w_{j_k}, w_l) \geq \delta_{i_k l}(y_{i_k}, y_l)$ for any $1 \leq k \leq m$ and l with $z_l = 0$. From (A6) and (A7), this immediately implies that for any $1 \leq k \leq n$,

$$\mathrm{r}_{j_k}(\boldsymbol{w}) = k + \sum_{l:z_l=0} \delta_{j_k l}(w_{j_k}, w_l) \geq k + \sum_{l:z_l=0} \delta_{i_k l}(y_{i_k}, y_l) = \mathrm{r}_{i_k}(\boldsymbol{y}). \tag{A8}$$

Third, we prove Lemma A3. From (A8) and the fact that $\phi(\cdot)$ is a monotone increasing function, we have

$$\sum_{i=1}^{n} z_i \phi\left(\mathrm{r}_i(\boldsymbol{w})\right) = \sum_{k=1}^{m} \phi\left(\mathrm{r}_{j_k}(\boldsymbol{w})\right) \geq \sum_{k=1}^{m} \phi\left(\mathrm{r}_{j_k}(\boldsymbol{y})\right) = \sum_{i=1}^{n} z_i \phi(\mathrm{r}_i(\boldsymbol{y})).$$

By the definition of $\boldsymbol{w}$, we then derive Lemma A3. □

Proof of Proposition 2. Below we consider the statistic $t_2(\cdot,\cdot)$ in (6), and assume that $\phi(\cdot)$ is a monotone increasing function and "first" or "last" method is used for ties. From Lemma A3, we can immediately know that $t_2(\cdot,\cdot)$ is effect increasing. Below we prove that it is also distribution free under exchangeable treatment assignment.

For any $\boldsymbol{y} \in \mathbb{R}^n$, because we use "first" or "last" method for ties, there must exists a permutations $\pi(\cdot)$ of $\{1, 2, \ldots, n\}$ such that $(r_{\pi(1)}(\boldsymbol{y}), r_{\pi(2)}(\boldsymbol{y}), \ldots, r_{\pi(n)}(\boldsymbol{y})) = (1, 2, \ldots, n)$, where π depends on both the outcome vector $\boldsymbol{y}$ and the original ordering of units. Consequently,

$$\sum_{i=1}^{n} Z_i \phi(\mathrm{r}_i(\boldsymbol{y})) = \sum_{i=1}^{n} Z_{\pi(i)} \phi(\mathrm{r}_{\pi(i)}(\boldsymbol{y})) = \sum_{i=1}^{n} Z_{\pi(i)} \phi(i). \tag{A9}$$

Because the ordering of units is independent of the treatment assignment, by Definition 1, conditional on $\pi(\cdot)$, (A9) follows the same distribution as $\sum_{i=1}^{n} Z_i \phi(i)$, which depends neither on the permutation $\pi(\cdot)$ nor the outcome vector $\boldsymbol{y}$. Therefore, the statistic $t_2(\cdot,\cdot)$ in (6) is distribution free, satisfying Definition 4.

From the above, Proposition 2 holds. □

Examples of statistics satisfying exactly one of Definitions 2–4. Below we construct statistics that satisfy only one of the three properties defined in Definitions 2–4.

First, we define $\tilde{t}_1(\boldsymbol{z}, \boldsymbol{y}) = y_{i(\boldsymbol{z})}$, where $i(\boldsymbol{z}) = \arg\min_{i: z_i = 1} i$. It is easy to show that $\tilde{t}_1(\cdot,\cdot)$ is effect increasing, and it is not distribution free even under a CRE. Below we give a counterexample to show that it is not diffential increasing. Define $\boldsymbol{a} = (1, 1, 0)^\top, \boldsymbol{z} = (0, 1, 1)^\top, \boldsymbol{y} = (0, 0, 0)^\top$, and $\boldsymbol{\eta} = (1, 2, 0)^\top \succcurlyeq \mathbf{0}$. Then we have $\tilde{t}_1(\boldsymbol{a}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - \tilde{t}_1(\boldsymbol{a}, \boldsymbol{y}) = (y_1 + a_1\eta_1) - y_1 = \eta_1 = 1$, and $\tilde{t}_1(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - \tilde{t}_1(\boldsymbol{z}, \boldsymbol{y}) = (y_2 + a_2\eta_2) - y_2 = \eta_2 = 2 > \tilde{t}_1(\boldsymbol{a}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - \tilde{t}_1(\boldsymbol{a}, \boldsymbol{y})$. Thus, $\tilde{t}_1(\cdot,\cdot)$ is not differential increasing.

Second, we define $\tilde{t}_2(\boldsymbol{z}, \boldsymbol{y}) = \sum_{i=1}^{n} z_i y_i - 2\sum_{i=1}^{n} y_i$. Using Proposition 1 with $t_1(\cdot,\cdot)$ in (5), ϕ_{1i} being identity function and ϕ_{0i} being zero function, we can know that $t_1(\boldsymbol{z}, \boldsymbol{y}) = \sum_{i=1}^{n} z_i y_i$ is differential increasing. Therefore, for any $\boldsymbol{z}, \boldsymbol{a} \in \mathcal{Z}$ and $\boldsymbol{y}, \boldsymbol{\eta} \in \mathbb{R}^n$ with $\boldsymbol{\eta} \succcurlyeq \mathbf{0}$,

$$\begin{aligned}
\tilde{t}_2(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - \tilde{t}_2(\boldsymbol{z}, \boldsymbol{y}) &= t_1(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - 2\sum_{i=1}^{n}(y_i + a_i\eta_i) - t_1(\boldsymbol{z}, \boldsymbol{y}) + 2\sum_{i=1}^{n} y_i \\
&= t_1(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - t_1(\boldsymbol{z}, \boldsymbol{y}) - 2\sum_{i=1}^{n}(y_i + a_i\eta_i) + 2\sum_{i=1}^{n} y_i \\
&\leq t_1(\boldsymbol{a}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - t_1(\boldsymbol{a}, \boldsymbol{y}) - 2\sum_{i=1}^{n}(y_i + a_i\eta_i) + 2\sum_{i=1}^{n} y_i \\
&= \tilde{t}_2(\boldsymbol{a}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - \tilde{t}_2(\boldsymbol{a}, \boldsymbol{y}).
\end{aligned}$$

Thus, $\tilde{t}_2(\cdot,\cdot)$ is differential increasing. It is easy to show that $\tilde{t}_2(\cdot,\cdot)$ is not distribution free even under a CRE. Below we give counterexample to show $\tilde{t}_2(\cdot,\cdot)$ is not effect increasing. Define $\boldsymbol{z} = (1,0)^\top, \boldsymbol{y} = (0,0)^\top$ and $\boldsymbol{\eta} = (1,0)^\top \succcurlyeq \mathbf{0}$. Then we have $\tilde{t}_2(\boldsymbol{z}, \boldsymbol{y}) = y_1 - 2(y_1 + y_2) = 0$, and $\tilde{t}_2(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{z} \circ \boldsymbol{\eta}) = (y_1 + \eta_1) - 2(y_1 + \eta_1 + y_2) = -1 < \tilde{t}_2(\boldsymbol{z}, \boldsymbol{y})$. Thus, $\tilde{t}_2(\cdot,\cdot)$ is not effect increasing.

Third, we define $\tilde{t}_3(\boldsymbol{z}, \boldsymbol{y}) = -\sum_{i=1}^{2} z_i \mathrm{r}_i(\boldsymbol{y})$ and consider a CRE where one unit is randomly assigned to treatment and the remaining one is randomly assigned to control. From Proposition 2, it is not hard to see that $\tilde{t}_3(\boldsymbol{z}, \boldsymbol{y})$ is not effect increasing but it is distribution free. Below we give a numerical example to show it is not differential increasing. Let $\boldsymbol{y} = (1,2)^\top$, $\boldsymbol{\eta} = (2,0)^\top$, $\boldsymbol{a} = (1,0)^\top$ and $\boldsymbol{z} = (0,1)^\top$. Then we have $\tilde{t}_3(\boldsymbol{z}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - \tilde{t}_3(\boldsymbol{z}, \boldsymbol{y}) = 1 > -1 = \tilde{t}_3(\boldsymbol{a}, \boldsymbol{y} + \boldsymbol{a} \circ \boldsymbol{\eta}) - \tilde{t}_3(\boldsymbol{a}, \boldsymbol{y})$. $\square$

A4. Proofs for broader justification of Fisher randomization test

A4.1. Proofs of Theorems 1, A1 and 6 and Corollaries 1 and A1

To prove Theorems 1 and A1, we need the following lemma.

Lemma A4. For any function $h(\cdot) : \mathcal{Z} \to \mathbb{R}$, let $\boldsymbol{Z}$ be the random treatment assignment vector, and $H(c) = \Pr\{h(\boldsymbol{Z}) \geq c\}$ be the tail probability of the random variable $h(\boldsymbol{Z})$. Then, $H(h(\boldsymbol{Z}))$ is stochastically larger than or equal to Unif$[0,1]$, in the sense that for any $\alpha \in (0,1)$, $\Pr\{H(h(\boldsymbol{Z})) \leq \alpha\} \leq \alpha$.

Proof of Lemma A4. By the property of probability measure, the tail probability function $H(\cdot)$ is decreasing and left-continuous. For any $\alpha \in (0,1)$, define $c_\alpha = \inf_c\{c : H(c) \leq \alpha\}$. Below we consider two cases in which $H(c_\alpha) \leq \alpha$ and $H(c_\alpha) > \alpha$, respectively.

First, we consider the case in which $H(c_\alpha) \leq \alpha$. By definition, we can know that $H(h(\boldsymbol{Z})) \leq \alpha$ is equivalent to $h(\boldsymbol{Z}) \geq c_\alpha$. This further implies that $\Pr\{H(h(\boldsymbol{Z})) \leq \alpha\} = \Pr\{h(\boldsymbol{Z}) \geq c_\alpha\} = H(c_\alpha) \leq \alpha$.

Second, we consider the case in which $H(c_\alpha) > \alpha$. By definition, we can know that $H(h(\boldsymbol{Z})) \leq \alpha$ is equivalent to $h(\boldsymbol{Z}) > c_\alpha$. This further implies that $\Pr\{H(h(\boldsymbol{Z})) \leq \alpha\} = \Pr\{h(\boldsymbol{Z}) > c_\alpha\} = \lim_{a \to 0+} \Pr\{h(\boldsymbol{Z}) \geq c_\alpha + a\} = \lim_{a \to 0+} H(c_\alpha + a) \leq \alpha$.

From the above, Lemma A4 holds. □

Proof of Theorem 1(a). Suppose the bounded null $H_{\preccurlyeq\boldsymbol{\delta}}$ in (7) holds, i.e., the true treatment effect satisfies $\boldsymbol{\tau} \preccurlyeq \boldsymbol{\delta}$. Then, by the definition in (2), the imputed potential outcomes satisfy

$$\boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(1) - \boldsymbol{Y}(1) = (\mathbf{1} - \boldsymbol{Z}) \circ (\boldsymbol{\delta} - \boldsymbol{\tau}) \succcurlyeq \mathbf{0} \quad \text{and} \quad \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0) - \boldsymbol{Y}(0) = \boldsymbol{Z} \circ (\boldsymbol{\tau} - \boldsymbol{\delta}) \preccurlyeq \mathbf{0}. \tag{A10}$$

This is because any actual treated value is smaller than or equal to the imputed, and any actual control value is larger than or equal to the imputed.

We first consider the case in which the test statistic is effect increasing. Similar to (2), for any $\boldsymbol{a} \in \mathcal{Z}$, the imputed control potential outcome if the observed treatment assignment was $\boldsymbol{a}$ would be $\boldsymbol{Y}_{\boldsymbol{a},\boldsymbol{\delta}}(0) = \boldsymbol{Y}(\boldsymbol{a}) - \boldsymbol{a} \circ \boldsymbol{\delta} = \boldsymbol{a} \circ \{\boldsymbol{Y}(1) - \boldsymbol{\delta}\} + (\mathbf{1} - \boldsymbol{a}) \circ \boldsymbol{Y}(0)$, and the difference between $\boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)$ and $\boldsymbol{Y}_{\boldsymbol{a},\boldsymbol{\delta}}(0)$ has the following equivalent forms:

$$\begin{aligned}
\boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0) - \boldsymbol{Y}_{\boldsymbol{a},\boldsymbol{\delta}}(0) &= \boldsymbol{a} \circ \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0) + (\mathbf{1} - \boldsymbol{a}) \circ \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0) - \boldsymbol{a} \circ \{\boldsymbol{Y}(1) - \boldsymbol{\delta}\} - (\mathbf{1} - \boldsymbol{a}) \circ \boldsymbol{Y}(0) \\
&= \boldsymbol{a} \circ \{\boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(1) - \boldsymbol{Y}(1)\} + (\mathbf{1} - \boldsymbol{a}) \circ \{\boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0) - \boldsymbol{Y}(0)\}
\end{aligned}$$

From (A10) and Definition 2, $t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) \geq t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{a},\boldsymbol{\delta}}(0))$. Thus, for any $c \in \mathbb{R}$, the imputed tail probability $G_{\boldsymbol{Z},\boldsymbol{\delta}}(c)$ in (3) of the test statistic can be bounded by

$$G_{\boldsymbol{Z},\boldsymbol{\delta}}(c) = \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) \geq c\right\} \geq \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{a},\boldsymbol{\delta}}(0)) \geq c\right\},$$

where the right hand side is actually the tail probability of the true randomization distribution of the test statistic $t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))$. From Lemma A4, we can derive that $p_{\boldsymbol{Z},\boldsymbol{\delta}} \equiv G_{\boldsymbol{Z},\boldsymbol{\delta}}(t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)))$ is stochastically larger than or equal to Unif$[0,1]$, i.e., it is a valid p-value for testing $H_{\preccurlyeq\boldsymbol{\delta}}$.

We then consider the case in which the test statistic is differential increasing. From (A10),

$$t(\boldsymbol{a}, \boldsymbol{Y}(0)) - t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) = t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0) + \boldsymbol{Z} \circ (\boldsymbol{\delta} - \boldsymbol{\tau})) - t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)). \tag{A11}$$

From Definition 3, the change of the statistic in (A11) is maximized at $\boldsymbol{a} = \boldsymbol{Z}$, which immediately implies that $t(\boldsymbol{Z}, \boldsymbol{Y}(0)) - t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) \geq t(\boldsymbol{a}, \boldsymbol{Y}(0)) - t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))$ for any $\boldsymbol{a} \in \mathcal{Z}$. Thus, $t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) - t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) \geq t(\boldsymbol{a}, \boldsymbol{Y}(0)) - t(\boldsymbol{Z}, \boldsymbol{Y}(0))$ for any $\boldsymbol{a} \in \mathcal{Z}$, and the randomization p-value $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ in (4) is bounded by

$$\begin{aligned}
p_{\boldsymbol{Z},\boldsymbol{\delta}} &= \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) \geq t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))\right\} \\
&\geq \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}(0)) \geq t(\boldsymbol{Z}, \boldsymbol{Y}(0))\right\},
\end{aligned}$$

which is actually the tail probability of the true randomization distribution of $t(\boldsymbol{Z}, \boldsymbol{Y}(0))$ evaluated at its realized value. From Lemma A4, we can derive that $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ is stochastically larger than or equal to Unif$[0,1]$, i.e., it is a valid p-value for testing $H_{\preccurlyeq\boldsymbol{\delta}}$.

From the above, Theorem 1(a) holds. As a side note, although both effect increasing and differential increasing test statistics can lead to valid randomization tests for bounded null hypotheses, their proofs are rather different, as shown above. Specifically, the randomization p-value $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ using effect increasing test statistic is bounded by the tail probability of $t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))$ evaluated at its realized value, while that using differential increasing test statistic is bounded by the tail probability of $t(\boldsymbol{Z}, \boldsymbol{Y}(0))$ evaluated at its realized value. The two bounds are generally different, although they are both stochastically larger than or equal to $\text{Unif}[0, 1]$. □

Proof of Theorem 1(b). For any $\boldsymbol{\delta}, \overline{\boldsymbol{\delta}} \in \mathbb{R}^n$ with $\boldsymbol{\delta} \preccurlyeq \overline{\boldsymbol{\delta}}$, by the definition in (2), the differences between the imputed treatment and control potential outcomes under nulls $\overline{\boldsymbol{\delta}}$ and $\boldsymbol{\delta}$ satisfy

$$\begin{aligned} \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(1) - \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(1) &= \boldsymbol{Y} + (\boldsymbol{1} - \boldsymbol{Z}) \circ \overline{\boldsymbol{\delta}} - \{\boldsymbol{Y} + (\boldsymbol{1} - \boldsymbol{Z}) \circ \boldsymbol{\delta}\} = (\boldsymbol{1} - \boldsymbol{Z}) \circ (\overline{\boldsymbol{\delta}} - \boldsymbol{\delta}) \succcurlyeq \boldsymbol{0}, \\ \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0) - \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0) &= \boldsymbol{Y} - \boldsymbol{Z} \circ \overline{\boldsymbol{\delta}} - \{\boldsymbol{Y} - \boldsymbol{Z} \circ \boldsymbol{\delta}\} = \boldsymbol{Z} \circ (\boldsymbol{\delta} - \overline{\boldsymbol{\delta}}) \preccurlyeq \boldsymbol{0}. \end{aligned} \tag{A12}$$

We first consider the case in which the test statistic is differential increasing. From (A12) and Definition 3, for any $\boldsymbol{\delta}, \overline{\boldsymbol{\delta}} \in \mathbb{R}^n$ with $\boldsymbol{\delta} \preccurlyeq \overline{\boldsymbol{\delta}}$ and any $\boldsymbol{a} \in \mathcal{Z}$,

$$\begin{aligned} t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) - t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0)) &= t\left(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0) + \boldsymbol{Z} \circ (\overline{\boldsymbol{\delta}} - \boldsymbol{\delta})\right) - t\left(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0)\right) \\ &\geq t\left(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0) + \boldsymbol{Z} \circ (\overline{\boldsymbol{\delta}} - \boldsymbol{\delta})\right) - t\left(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0)\right) \\ &= t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) - t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0)). \end{aligned}$$

Consequently, for any $\boldsymbol{a} \in \mathcal{Z}$,

$$t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0)) - t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0)) \geq t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) - t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)).$$

By the definition in (4),

$$\begin{aligned} p_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}} &= \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0)) \geq t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0))\right\} \\ &\geq \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) \geq t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))\right\} = p_{\boldsymbol{Z},\boldsymbol{\delta}}. \end{aligned}$$

We then consider the case in which the test statistic is effect increasing and distribution free. From (A12) and Definition 2, for any $\boldsymbol{\delta}, \overline{\boldsymbol{\delta}} \in \mathbb{R}^n$ with $\boldsymbol{\delta} \preccurlyeq \overline{\boldsymbol{\delta}}$,

$$t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) = t\left(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0) + \boldsymbol{Z} \circ (\overline{\boldsymbol{\delta}} - \boldsymbol{\delta})\right) \geq t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0)). \tag{A13}$$

From (3) and Definition 4, we can know that

$$G_{\boldsymbol{Z},\boldsymbol{\delta}}(c) \equiv \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)) \geq c\right\} = G_0(c), \tag{A14}$$

a function that does not depend on $\boldsymbol{Z}$ or $\boldsymbol{\delta}$. Moreover, $G_0(c)$ is decreasing in c. From (A13) and

(A14), by the definition in (4), we then have

$$
\begin{aligned}
p_{\boldsymbol{Z},\boldsymbol{\delta}} &= G_{\boldsymbol{Z},\boldsymbol{\delta}}\left\{t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))\right\} = G_0\left\{t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))\right\} \\
&\leq G_0\left\{t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0))\right\} = G_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}\left\{t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0))\right\} = p_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}.
\end{aligned}
$$

From the above, Theorem 1(b) holds. □

Proof of Theorem A1. Suppose Theorem A1(b) holds. Then when the bounded null $H_{\preccurlyeq\boldsymbol{\delta}}$ in (7) holds, i.e., $\boldsymbol{\tau} \preccurlyeq \boldsymbol{\delta}$, we must have $\tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}} \geq \tilde{p}_{\boldsymbol{Z},\boldsymbol{\tau}}$, which is stochastically larger than or equal to Unif$[0,1]$ by the validity of Fisher randomization test for sharp null hypothesis $H_{\boldsymbol{\delta}} : \boldsymbol{\delta} = \boldsymbol{\tau}$. Therefore, Theorem A1(b) implies A1(a), and, to prove Theorem A1, it suffices to prove Theorem A1(b). Below we prove Theorem A1(b).

Let $\boldsymbol{\delta}, \overline{\boldsymbol{\delta}} \in \mathbb{R}^n$ be two constant vectors satisfying $\boldsymbol{\delta} \preccurlyeq \overline{\boldsymbol{\delta}}$. For any $\boldsymbol{a} \in \mathcal{Z}$, the corresponding observed outcomes from the imputed potential outcomes under nulls $\overline{\boldsymbol{\delta}}$ and $\boldsymbol{\delta}$ satisfy

$$
\begin{aligned}
\boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(\boldsymbol{a}) - \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(\boldsymbol{a}) &= \boldsymbol{a} \circ \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(1) + (\boldsymbol{1} - \boldsymbol{a}) \circ \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0) - \left\{\boldsymbol{a} \circ \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(1) + (\boldsymbol{1} - \boldsymbol{a}) \circ \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)\right\} \\
&= \boldsymbol{a} \circ \left\{\boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(1) - \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(1)\right\} + (\boldsymbol{1} - \boldsymbol{a}) \circ \left\{\boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(0) - \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0)\right\}.
\end{aligned}
$$

From (A12) and Definition 2, for any effect increasing statistic $t(\cdot,\cdot)$ and any $\boldsymbol{a} \in \mathcal{Z}$, $t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(\boldsymbol{a})) \geq t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(\boldsymbol{a}))$. By the definition in (A2), we then have

$$
\begin{aligned}
\tilde{p}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}} &= \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}}(\boldsymbol{a})) \geq t(\boldsymbol{Z}, \boldsymbol{Y})\right\} \\
&\geq \sum_{\boldsymbol{a} \in \mathcal{Z}} \Pr(\boldsymbol{A} = \boldsymbol{a}) \mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(\boldsymbol{a})) \geq t(\boldsymbol{Z}, \boldsymbol{Y})\right\} = \tilde{p}_{\boldsymbol{Z},\boldsymbol{\delta}}.
\end{aligned}
$$

Therefore, Theorem A1(b) holds. □

Proof of Corollary 1. We first prove (a) in Corollary 1. By definition, the coverage probability of the set $\{c : p_{\boldsymbol{Z},c\boldsymbol{1}} > \alpha, c \in \mathbb{R}\}$ has the following equivalent forms:

$$
\Pr\left(\tau_{\max} \in \{c : p_{\boldsymbol{Z},c\boldsymbol{1}} > \alpha, c \in \mathbb{R}\}\right) = \Pr\left(p_{\boldsymbol{Z},\tau_{\max}\boldsymbol{1}} > \alpha\right) = 1 - \Pr\left(p_{\boldsymbol{Z},\tau_{\max}\boldsymbol{1}} \leq \alpha\right). \tag{A15}
$$

Because the test statistic is either differential increasing or effect increasing, from Theorem 1(a), $p_{\boldsymbol{Z},\tau_{\max}\boldsymbol{1}}$ is a valid p-value for testing the bounded null $H_{\tau_{\max}\boldsymbol{1}}$. Because the null $H_{\tau_{\max}\boldsymbol{1}}$ holds by definition, we have $\Pr(p_{\boldsymbol{Z},\tau_{\max}\boldsymbol{1}} \leq \alpha) \leq \alpha$, and thus the coverage probability (A15) is larger than or equal to $1 - \alpha$. Therefore, $\{c : p_{\boldsymbol{Z},c\boldsymbol{1}} > \alpha, c \in \mathbb{R}\}$ is a $1 - \alpha$ confidence set for $\tau_{\max}$.

We then prove (b) in Corollary 1. Because the test statistic is either differential increasing, or both effect increasing and distribution free, from Theorem 1(b), $p_{\boldsymbol{Z},c\boldsymbol{1}}$ is increasing in c, which implies that the confidence set must have the form of $(\underline{c}, \infty)$ or $[\underline{c}, \infty)$ with $\underline{c} = \inf\{c : p_{\boldsymbol{Z},c\boldsymbol{1}} > \alpha, c \in \mathbb{R}\}$.

From the above, Corollary 1 holds. □

Proof of Corollary A1. Corollary A1 follows from Theorem A1, and its proof is almost the same as that for Corollary 1. Thus, we omit its proof here. □

Proof of Theorem 6. First, the coverage probability of the interval $[\max\{\hat\tau^L_{\max} - \hat\tau^U_{\min}, 0\}, \infty)$ is

$$\begin{aligned}
&\Pr\left(\tau_{\max} - \tau_{\min} \geq \max\{\hat\tau^L_{\max} - \hat\tau^U_{\min}, 0\}\right) \\
&= \Pr\left(\tau_{\max} - \tau_{\min} \geq \hat\tau^L_{\max} - \hat\tau^U_{\min}\right) \geq \Pr\left(\tau_{\max} \geq \hat\tau^L_{\max}, \tau_{\min} \leq \hat\tau^U_{\min}\right) \\
&= 1 - \Pr\left(\tau_{\max} < \hat\tau^L_{\max} \text{ or } \tau_{\min} > \hat\tau^U_{\min}\right) \geq 1 - \Pr\left(\tau_{\max} < \hat\tau^L_{\max}\right) - \Pr\left(\tau_{\min} > \hat\tau^U_{\min}\right) \\
&\geq 1 - \alpha/2 - \alpha/2 = 1 - \alpha,
\end{aligned}$$

where the last equality holds because $[\hat\tau^L_{\max}, \infty)$ and $(-\infty, \hat\tau^U_{\min}]$ are $1-\alpha/2$ confidence intervals for $\tau_{\max}$ and $\tau_{\min}$, respectively. Thus, (a) in Theorem 6 holds.

Second, suppose the null hypothesis of constant treatment effect is true. Then we must have $\tau_{\max} - \tau_{\min} = 0$. From (a) in Theorem 6,

$$\begin{aligned}
\Pr\left(\hat\tau^L_{\max} - \hat\tau^U_{\min} > 0\right) &= \Pr\left(\tau_{\max} - \tau_{\min} \notin [\hat\tau^L_{\max} - \hat\tau^U_{\min}, \infty)\right) \\
&= 1 - \Pr\left(\tau_{\max} - \tau_{\min} \in [\hat\tau^L_{\max} - \hat\tau^U_{\min}, \infty)\right) \leq 1 - (1-\alpha) \\
&= \alpha,
\end{aligned}$$

which implies that the probability of type-I error is at most α. Thus, (b) in Theorem 6 holds.

From the above, Theorem 6 holds. □

A4.2. Additional comments on randomization p-values

Example of a non-monotone p-value that is valid for bounded null. Here we present a numerical example showing that a valid p-value for testing the bounded null may not have the monotonicity property. We consider a CRE with 3 units, where 2 units are assigned to treatment group and the remaining 1 is assigned to control group. Suppose that the observed treatment assignment vector is $\boldsymbol{Z} = (1,1,0)^\top$ and the observed outcome vector is $\boldsymbol{Y} = (0,0,0)^\top$. We consider using the randomization p-value $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ in (4) with test statistic $t(\boldsymbol{z},\boldsymbol{y}) = y_{i(\boldsymbol{z})}$, where $i(\boldsymbol{z}) = \arg\min_{i:z_i=1} i$. From the examples of statistics discussed at the end of Appendix A3, the statistic $t(\boldsymbol{z},\boldsymbol{y}) = y_{i(\boldsymbol{z})}$ is effect increasing. From Theorem 1(a), the randomization p-value $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ is valid for testing the bounded null $H_{\preccurlyeq\boldsymbol{\delta}}$. However, the randomization p-value $p_{\boldsymbol{Z},\boldsymbol{\delta}}$ is not monotone in $\boldsymbol{\delta}$. For example, with $\underline{\boldsymbol{\delta}} = (-1,0,0)^\top \preccurlyeq \boldsymbol{\delta} = (0,0,0)^\top \preccurlyeq \overline{\boldsymbol{\delta}} = (0,1,0)^\top$, we have $p_{\boldsymbol{Z},\underline{\boldsymbol{\delta}}} = 2/3 < p_{\boldsymbol{Z},\boldsymbol{\delta}} = 1 > p_{\boldsymbol{Z},\overline{\boldsymbol{\delta}}} = 2/3$. □

Comment on the equivalence between randomization p-values $p_{\boldsymbol{Z},c\boldsymbol{1}}$ and $\tilde p_{\boldsymbol{Z},c\boldsymbol{1}}$. We are going to show that the two randomization p-values $p_{\boldsymbol{Z},c\boldsymbol{1}}$ in (A2) and $\tilde p_{\boldsymbol{Z},c\boldsymbol{1}}$ in (4) are equivalent for testing the null hypothesis $H_{c\boldsymbol{1}}$ of constant effect c for any $c \in \mathbb{R}$, if both of them use difference-in-means as the test statistics. For any $c \in \mathbb{R}$, the difference-in-means statistics for the two randomization p-values evaluated at assignment $\boldsymbol{a} \in \mathcal{Z}$ are, respectively,

$$t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0)) = \frac{1}{m}\boldsymbol{a}^\top \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0) - \frac{1}{n-m}(\mathbf{1}-\boldsymbol{a})^\top \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0),$$

and

$$\begin{aligned}
t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(\boldsymbol{a})) &= \frac{1}{m}\boldsymbol{a}^\top \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(1) - \frac{1}{n-m}(\mathbf{1}-\boldsymbol{a})^\top \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0) \\
&= \frac{1}{m}\boldsymbol{a}^\top \left\{\boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0) + c\mathbf{1}\right\} - \frac{1}{n-m}(\mathbf{1}-\boldsymbol{a})^\top \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0) \\
&= \frac{1}{m}\boldsymbol{a}^\top \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0) - \frac{1}{n-m}(\mathbf{1}-\boldsymbol{a})^\top \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0) + \frac{1}{m}\boldsymbol{a}^\top(c\mathbf{1}) \\
&= t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0)) + c.
\end{aligned}$$

Therefore, the randomization p-values $p_{\boldsymbol{Z},c\mathbf{1}}$ and $\tilde{p}_{\boldsymbol{Z},c\mathbf{1}}$ in (4) and (A2) satisfy

$$\begin{aligned}
p_{\boldsymbol{Z},c\mathbf{1}} &= \sum_{\boldsymbol{a}\in\mathcal{Z}} \Pr(\boldsymbol{A}=\boldsymbol{a})\mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0)) \geq t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(0))\right\} \\
&= \sum_{\boldsymbol{a}\in\mathcal{Z}} \Pr(\boldsymbol{A}=\boldsymbol{a})\mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(\boldsymbol{a})) - c \geq t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(\boldsymbol{Z})) - c\right\} \\
&= \sum_{\boldsymbol{a}\in\mathcal{Z}} \Pr(\boldsymbol{A}=\boldsymbol{a})\mathbb{1}\left\{t(\boldsymbol{a}, \boldsymbol{Y}_{\boldsymbol{Z},c\mathbf{1}}(\boldsymbol{a})) \geq t(\boldsymbol{Z}, \boldsymbol{Y})\right\} \\
&= \tilde{p}_{\boldsymbol{Z},c\mathbf{1}}.
\end{aligned}$$

Therefore, for any $c \in \mathbb{R}$, the randomization p-values $p_{\boldsymbol{Z},c\mathbf{1}}$ and $\tilde{p}_{\boldsymbol{Z},c\mathbf{1}}$ with the difference-in-means statistics are equivalent for testing the null hypothesis $H_{c\mathbf{1}}$ of constant effect c. □

A5. Proofs for randomization inference on quantiles of individual effects

Proof of Theorem 2. From the property of usual randomization test, $p_{\boldsymbol{Z},\boldsymbol{\tau}}$ is a valid p-value, i.e., $\Pr(p_{\boldsymbol{Z},\boldsymbol{\tau}} \leq \alpha) \leq \alpha$ for any $\alpha \in [0,1]$. When the null hypothesis $\boldsymbol{\tau} \in \mathcal{H}$ is true, for any $\alpha \in [0,1]$, we have $\Pr\left(\sup_{\boldsymbol{\delta}\in\mathcal{H}} p_{\boldsymbol{Z},\boldsymbol{\delta}} \leq \alpha\right) \leq \Pr(p_{\boldsymbol{Z},\boldsymbol{\tau}} \leq \alpha) \leq \alpha$. Thus, $\sup_{\boldsymbol{\delta}\in\mathcal{H}} p_{\boldsymbol{Z},\boldsymbol{\delta}}$, is a valid p-value for testing the null hypothesis of $\boldsymbol{\tau} \in \mathcal{H}$. Consequently, the upper bound on the right hand side of (10) is also a valid p-value for testing the null of $\boldsymbol{\tau} \in \mathcal{H}$. Therefore, Theorem 2 holds. □

To prove Theorem 3, we need the following three lemmas.

Lemma A5. When the treatment assignment $\boldsymbol{Z}$ is exchangeable as in Definition 1 and independent of the ordering of the units, the rank score statistic in Definition 5 using "first" method for ties is both effect increasing as in Definition 2 and distribution free as in Definition 4.

Proof of Lemma A5. Lemma A5 follows immediately from Proposition 2. □

Lemma A6. For any $1 \leq k \leq n$, $\boldsymbol{z} \in \{0,1\}^n$ and $\boldsymbol{y} \in \mathbb{R}^n$, let $m = \sum_{i=1}^n z_i$, and $\mathcal{I}_k$ be the set of indices for the largest $\min(n-k, m)$ coordinates of $\boldsymbol{y}$ with corresponding z_i's being 1, i.e.,

$$\mathcal{I}_k \subset \{i : z_i = 1, 1 \leq i \leq n\}, \ \ |\mathcal{I}_k| = \min(n-k, m), \ \ \inf_{i \in \mathcal{I}_k} \mathrm{r}_i(\boldsymbol{y}) > \sup_{i: i \notin \mathcal{I}_k, z_i = 1} \mathrm{r}_i(\boldsymbol{y}),$$

where $\mathrm{r}(\cdot)$ uses the "first" method for ties. Then for any rank score statistic $t(\cdot,\cdot)$, we have $\inf_{\boldsymbol{\delta} \in \mathcal{H}_{k,0}} t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\delta}) = t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi})$, where $\boldsymbol{\xi} = (\xi_1, \xi_2, \ldots, \xi_n)'$ is defined as $\xi_i = \infty$ if $i \in \mathcal{I}_k$ and 0 otherwise.

Proof of Lemma A6. To ease the description, define $l = \min(n-k, m)$ and $\mathcal{T} = \{i : z_i = 1, 1 \leq i \leq n\}$. Let $r_1 < r_2 < \ldots < r_m$ be the ranks of y_i's with indices in $\mathcal{T}$, i.e., $\{r_1, r_2, \ldots, r_m\} = \{\mathrm{r}_i(\boldsymbol{y}) : i \in \mathcal{T}\}$.

First, we show that for any $\boldsymbol{\delta} \in \mathcal{H}_{k,0}$, there exists a set $\mathcal{J}_k \subset \mathcal{T}$ of cardinality l such that $t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\delta}) \geq t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta})$, where $\boldsymbol{\eta} = (\eta_1, \eta_2, \ldots, \eta_n)'$ satisfies $\eta_i = \infty$ if $i \in \mathcal{J}_k$ and zero otherwise. For any $\boldsymbol{\delta} \in \mathcal{H}_{k,0}$, let $\mathcal{J}_k$ be the set of indices for the largest l coordinates of $\boldsymbol{\delta}$ with indices in $\mathcal{T}$, i.e., $\mathcal{J}_k \subset \mathcal{T}$, $|\mathcal{J}_k| = l$ and $\min_{i \in \mathcal{J}_k} \delta_i > \max_{i: i \notin \mathcal{J}_k, z_i = 1} \delta_i$. Then by the definition of $\mathcal{H}_{k,0}$ in Section 5.2, for any $i \in \mathcal{T} \setminus \mathcal{J}_k$, we must have $\delta_i \leq 0$. Define a vector $\boldsymbol{\eta} = (\eta_1, \eta_2, \ldots, \eta_n)'$ with $\eta_i = \infty$ if $i \in \mathcal{J}_k$ and 0 otherwise. For any $i \in \mathcal{T}$, we have $\delta_i \leq \eta_i$ and thus $y_i - z_i\delta_i \geq y_i - z_i\eta_i$. For any $i \notin \mathcal{T}$, we have $y_i - z_i\delta_i = y_i = y_i - z_i\eta_i$. Because the test statistic is effect increasing, from Definition 2, we have $t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\delta}) \geq t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta})$.

Second, for any $\mathcal{J}_k \subset \mathcal{T}$ of cardinality l, we calculate the value of $t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta})$, where $\boldsymbol{\eta} = (\eta_1, \eta_2, \ldots, \eta_n)'$ satisfies $\eta_i = \infty$ if $i \in \mathcal{J}_k$ and zero otherwise. Recall that $\{r_1 < r_2 < \ldots < r_m\} = \{\mathrm{r}_i(\boldsymbol{y}) : i \in \mathcal{T}\}$. Let $r_{j_1} < r_{j_2} < \ldots, r_{j_{m-l}}$ be the ranks of the y_i's with indices in $\mathcal{T} \setminus \mathcal{J}_k$, and $r_{j_{m-l+1}} < r_{j_{m-l+2}} < \ldots < r_{j_m}$ be the ranks of the y_i's with indices in $\mathcal{J}_k$, i.e.,

$$\{\mathrm{r}_i(\boldsymbol{y}) : i \in \mathcal{T} \setminus \mathcal{J}_k\} = \{r_{j_1}, r_{j_2}, \ldots, r_{j_{m-l}}\}, \quad \{\mathrm{r}_i(\boldsymbol{y}) : i \in \mathcal{J}_k\} = \{r_{j_{m-l+1}}, r_{j_{m-l+2}}, \ldots, r_{j_m}\}$$

where $\{j_1, j_2, \ldots, j_m\}$ is a permutation of $\{1, 2, \ldots, m\}$. By the definition of $\mathcal{J}_k$ and $\boldsymbol{\eta}$, $y_i - z_i\eta_i = -\infty$ for $i \in \mathcal{J}_k$, and thus the ranks of $y_i - z_i\eta_i$'s with $i \in \mathcal{J}_k$ must become $\{1, 2, \ldots, l\}$. For each $i \in \mathcal{T} \setminus \mathcal{J}_k$ with $\mathrm{r}_i(\boldsymbol{y}) = r_{j_p}$ for some $1 \leq p \leq m - l$, there are $(m - j_p)$ coordinates of $\boldsymbol{y}$ with indices in $\mathcal{T}$ having larger ranks than y_i. However, for coordinates of $\boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}$ with indices in $\mathcal{T}$, there are only $m - l - p$ of them ranked higher than $y_i - z_i\eta_i = y_i$. Note that $y_j - z_j\eta_j = y_j$ for $j \notin \mathcal{T}$. These imply that the rank of $y_i - z_i\eta_i$ increases by $(m - j_p) - (m - l - p) = l + p - j_p$ compared to the rank of the corresponding y_i. Consequently,

$$\begin{aligned} &\{\mathrm{r}_i(\boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) : i \in \mathcal{J}_k\} = \{1, 2, \ldots, l\}, \qquad\qquad \text{(A16)} \\ &\{\mathrm{r}_i(\boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) : i \in \mathcal{T} \setminus \mathcal{J}_k\} = \{r_{j_1} + l + 1 - j_1,\ r_{j_2} + l + 2 - j_2,\ \ldots,\ r_{j_{m-l}} + l + (m-l) - j_{m-l}\}. \end{aligned}$$

Therefore, the value of the statistic $t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta})$ is

$$t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) = \sum_{i \in \mathcal{T}} \phi \left\{ \mathrm{r}_i(\boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) \right\} = \sum_{i \in \mathcal{J}_k} \phi \left\{ \mathrm{r}_i(\boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) \right\} + \sum_{i \in \mathcal{T} \setminus \mathcal{J}_k} \phi \left\{ \mathrm{r}_i(\boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) \right\}$$
$$= \sum_{i=1}^{l} \phi(i) + \sum_{p=1}^{m-l} \phi \left(r_{j_p} + l + p - j_p \right). \tag{A17}$$

Third, we calculate the value of $t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi})$, which is a special case of (A17) with $\mathcal{J}_k$ and $\boldsymbol{\eta}$ being $\mathcal{I}_k$ and $\boldsymbol{\xi}$. By definition, we can know that the ranks of y_i's with indices in $\mathcal{T} \setminus \mathcal{I}_k$ becomes $\{r_1, r_2, \ldots, r_{m-l}\}$, and the ranks of y_i's with indices in $\mathcal{I}_k$ must be $\{r_{m-l+1}, r_{m-l+2}, \ldots, r_m\}$, i.e.,

$$\{\mathrm{r}_i(\boldsymbol{y}) : i \in \mathcal{T} \setminus \mathcal{I}_k\} = \{r_1, r_2, \ldots, r_{m-l}\}, \quad \{\mathrm{r}_i(\boldsymbol{y}) : i \in \mathcal{I}_k\} = \{r_{m-l+1}, r_{m-l+2}, \ldots, r_m\}.$$

Using (A17) in the special case with $\mathcal{J}_k = \mathcal{I}_k$ and $\boldsymbol{\eta} = \boldsymbol{\xi}$, we have $(j_1, j_2, \ldots, j_{m-l}) = (1, 2, \ldots, m - l)$, and thus

$$t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi}) = \sum_{i=1}^{l} \phi(i) + \sum_{p=1}^{m-l} \phi \left(r_p + l + p - p \right) = \sum_{i=1}^{l} \phi(i) + \sum_{p=1}^{m-l} \phi \left(r_p + l \right). \tag{A18}$$

Fourth, we prove that, for any $\mathcal{J}_k \subset \mathcal{T}$ of cardinality l and $\boldsymbol{\eta} = (\eta_1, \eta_2, \ldots, \eta_n)'$ with $\eta_i = \infty$ if $i \in \mathcal{J}_k$ and zero otherwise, $t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) \geq t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi})$. From (A17) and (A18),

$$t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) - t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi}) = \sum_{i=1}^{l} \phi(i) + \sum_{p=1}^{m-l} \phi \left(r_{j_p} + l + p - j_p \right) - \sum_{i=1}^{l} \phi(i) - \sum_{p=1}^{m-l} \phi \left(r_p + l \right)$$
$$= \sum_{p=1}^{m-l} \left\{ \phi \left(r_{j_p} + l + p - j_p \right) - \phi \left(r_p + l \right) \right\}.$$

By the definition of $(r_{j_1}, r_{j_2}, \ldots, r_{j_p})$, for any $1 \leq p \leq m - l$, we must have $r_{j_p} \geq r_p$, or equivalently $j_p \geq p$. This further implies that for $1 \leq p \leq m - l$, $r_{j_p} - r_p = \sum_{i=p+1}^{j_p} (r_i - r_{i-1}) \geq \sum_{i=p+1}^{j_p} 1 = j_p - p$. Consequently, $r_{j_p} + l + p - j_p \geq r_p + l$. Therefore, we must have

$$t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\eta}) - t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi}) = \sum_{p=1}^{m-l} \left\{ \phi \left(r_{j_p} + l + p - j_p \right) - \phi \left(r_p + l \right) \right\} \geq 0.$$

From the above, we can derive Lemma A6. □

Lemma A7. For any $1 \leq k \leq n$, $\boldsymbol{z} \in \{0, 1\}^n$ and $\boldsymbol{y} \in \mathbb{R}^n$, let $m = \sum_{i=1}^{n} z_i$, and $\mathcal{I}_k$ be the set of indices for the largest $\min(n - k, m)$ coordinates of $\boldsymbol{y}$ with corresponding z_i's being 1, i.e.,

$$\mathcal{I}_k \subset \{i : z_i = 1, 1 \leq i \leq n\}, \quad |\mathcal{I}_k| = \min(n - k, m), \quad \inf_{i \in \mathcal{I}_k} \mathrm{r}_i(\boldsymbol{y}) > \sup_{i : i \notin \mathcal{I}_k, z_i = 1} \mathrm{r}_i(\boldsymbol{y}),$$

where $\mathrm{r}(\cdot)$ uses the "first" method for ties. Then for any rank score statistic $t(\cdot, \cdot)$ and any constant $c \in \mathbb{R}$,

(a) $\inf_{\delta \in \mathcal{H}_{k,c}} t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\delta}) = t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi}_{k,c})$, where $\boldsymbol{\xi} = (\xi_{1k,c}, \xi_{2k,c}, \ldots, \xi_{nk,c})'$ is defined as $\xi_{ik,c} = \infty$ if $i \in \mathcal{I}_k$ and c otherwise.

(b) $t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi}_{k,c}) = t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\zeta}_{k,c})$, where $\boldsymbol{\xi} = (\xi_{1k,c}, \xi_{2k,c}, \ldots, \xi_{nk,c})'$ is defined as $\xi_{ik,c} = \Delta$ if $i \in \mathcal{I}_k$ and c otherwise, and Δ is a constant larger than $\max_{j:z_j=1} y_j - \min_{j:z_j=0} y_j$

Proof of Lemma A7. We first prove (a) in Lemma A7. By definition, $\mathcal{H}_{k,c} = \mathcal{H}_{k,0} + c$. This implies that

$$\inf_{\delta \in \mathcal{H}_{k,c}} t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\delta}) = \inf_{\boldsymbol{\eta} \in \mathcal{H}_{k,0}} t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ (\boldsymbol{\eta} + c)) = \inf_{\boldsymbol{\eta} \in \mathcal{H}_{k,0}} t(\boldsymbol{z}, (\boldsymbol{y} - \boldsymbol{z} \circ c) - \boldsymbol{z} \circ \boldsymbol{\eta}).$$

Using Lemma A6, we then have

$$\begin{aligned}
\inf_{\delta \in \mathcal{H}_{k,c}} t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\delta}) &= \inf_{\boldsymbol{\eta} \in \mathcal{H}_{k,0}} t(\boldsymbol{z}, (\boldsymbol{y} - \boldsymbol{z} \circ c) - \boldsymbol{z} \circ \boldsymbol{\eta}) = t(\boldsymbol{z}, (\boldsymbol{y} - \boldsymbol{z} \circ c) - \boldsymbol{z} \circ \boldsymbol{\zeta}) \\
&= t(\boldsymbol{z}, \boldsymbol{y} - \boldsymbol{z} \circ (\boldsymbol{\zeta} + c)),
\end{aligned}$$

where $\boldsymbol{\zeta} = (\zeta_1, \zeta_2, \ldots, \zeta_n)'$ is defined as $\zeta_i = \infty$ if $i \in \mathcal{I}_k$ and 0 otherwise. Note that $\boldsymbol{\xi}_{k,c} = \boldsymbol{\zeta} + c$. We can then derive Lemma A7.

We then prove (b) in Lemma A7. Let $\boldsymbol{\beta} = \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\xi}_{k,c}$ and $\boldsymbol{\gamma} = \boldsymbol{y} - \boldsymbol{z} \circ \boldsymbol{\zeta}_{k,c}$. We can verify that $t(\boldsymbol{z}, \boldsymbol{\beta}) = \sum_{i=1}^n \phi(i) - \sum_{i=1}^n (1 - z_i)\phi(\mathrm{r}_i(\boldsymbol{\beta}))$ and $t(\boldsymbol{z}, \boldsymbol{\gamma}) = \sum_{i=1}^n \phi(i) - \sum_{i=1}^n (1 - z_i)\phi(\mathrm{r}_i(\boldsymbol{\gamma}))$. Thus, to prove (b), it suffices to prove that $\mathrm{r}_i(\boldsymbol{\beta}) = \mathrm{r}_i(\boldsymbol{\gamma})$ for any i with $z_i = 0$. Let $\mathcal{T} = \{l : z_l = 1\}$. Given $i \notin \mathcal{T}$, we count the number of coordinates of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ that have smaller ranks then the corresponding ith coordinates, respectively. First, for any $j \in \mathcal{I}_k$, by construction, $\beta_j = y_j - \infty = -\infty < y_i = \beta_i$ and $\gamma_j = y_j - \Delta < \min_{l:z_l=0} y_l < y_i = \gamma_i$. These imply that $\mathrm{r}_j(\boldsymbol{\beta}) < \mathrm{r}_i(\boldsymbol{\beta})$ and $\mathrm{r}_j(\boldsymbol{\gamma}) < \mathrm{r}_i(\boldsymbol{\gamma})$ for $j \in \mathcal{I}_k$. Second, for any $j \in \mathcal{T} \setminus \mathcal{I}_k$, by construction, $\beta_j = y_j - z_j c = \gamma_j$, and $\beta_i = y_i = \gamma_i$. Thus, $\mathrm{r}_j(\boldsymbol{\beta}) < \mathrm{r}_i(\boldsymbol{\beta})$ if and only if $\mathrm{r}_j(\boldsymbol{\gamma}) < \mathrm{r}_i(\boldsymbol{\gamma})$ for $j \in \mathcal{T} \setminus \mathcal{I}_k$. From the above, we must have $\mathrm{r}_i(\boldsymbol{\beta}) = \mathrm{r}_i(\boldsymbol{\gamma})$ for $i \notin \mathcal{T}$. Thus, (b) in Lemma A7 holds.

Therefore, Lemma A7 holds. □

Proof of Theorem 3. First, from Theorem 2, $p_{\boldsymbol{Z},k,c} \equiv \sup_{\delta \in \mathcal{H}_{k,c}} p_{\boldsymbol{Z},\boldsymbol{\delta}}$ is a valid p-value for testing the null hypothesis $H_{k,c}$ in (11). Second, from (10) and Lemma A7, $t(\boldsymbol{Z}, \boldsymbol{Y}_{\boldsymbol{Z},\boldsymbol{\delta}}(0))$ can achieve its infimum at some $\boldsymbol{\delta} \in \mathcal{H}_{k,c}$, and thus the equivalent forms of $p_{\boldsymbol{Z},k,c}$ in (13) holds. Therefore, Theorem 3 holds. □

Proof of Theorem 4. First, because $\mathcal{H}_{k,c} \subset \mathcal{H}_{k,\bar{c}}$ for any $c \leq \bar{c}$, from (13), we have

$$p_{\boldsymbol{Z},k,c} = \sup_{\delta \in \mathcal{H}_{k,c}} p_{\boldsymbol{Z},\boldsymbol{\delta}} \leq \sup_{\delta \in \mathcal{H}_{k,\bar{c}}} p_{\boldsymbol{Z},\boldsymbol{\delta}} = p_{\boldsymbol{Z},k,\bar{c}}.$$

Therefore, the p-value $p_{\boldsymbol{z},k,c}$, viewed as a function of c, is monotone increasing for any fixed $\boldsymbol{z}$ and k.

Second, the coverage probability of the set $\{c : p_{\boldsymbol{Z},k,c} > \alpha, c \in \mathbb{R}\}$ is

$$\Pr\left(\tau_{(k)} \in \{c : p_{\boldsymbol{Z},k,c} > \alpha, c \in \mathbb{R}\}\right) = \Pr\left(p_{\boldsymbol{Z},k,\tau_{(k)}} > \alpha\right) = 1 - \Pr\left(p_{\boldsymbol{Z},k,\tau_{(k)}} \leq \alpha\right) \geq 1 - \alpha,$$

where the last inequality holds because $p_{\boldsymbol{Z},k,\tau_{(k)}}$ is a valid p-value for testing the null hypothesis of $\tau_{(k)} \leq \tau_{(k)}$ that always holds by definition. Therefore, $\{c : p_{\boldsymbol{Z},k,c} > \alpha, c \in \mathbb{R}\}$ is a $1-\alpha$ confidence set for $\tau_{(k)}$.

Third, because $p_{\boldsymbol{z},k,c}$ is increasing in c, the confidence set $\{c : p_{\boldsymbol{Z},k,c} > \alpha, c \in \mathbb{R}\}$ must have the form of $(\underline{c}, \infty)$ or $[\underline{c}, \infty)$ with $\underline{c} = \inf\{c : p_{\boldsymbol{Z},k,c} > \alpha, c \in \mathbb{R}\}$.

From the above, Theorem 4 holds. □

Proof of Corollary 2. First, (a) in Corollary 2 follows directly from Theorem 3. Second, for any $1 \leq \underline{k} \leq k \leq n$, because $\mathcal{H}_{k,c} \subset \mathcal{H}_{\underline{k},c}$, we have $p_{\boldsymbol{Z},k,c} \equiv \sup_{\boldsymbol{\delta} \in \mathcal{H}_{k,c}} p_{\boldsymbol{Z},\boldsymbol{\delta}} \leq \sup_{\boldsymbol{\delta} \in \mathcal{H}_{\underline{k},c}} p_{\boldsymbol{Z},\boldsymbol{\delta}} \equiv p_{\boldsymbol{Z},\underline{k},c}$. This implies (b) in Corollary 2. Third, the coverage probability of the set $\{n-k : p_{\boldsymbol{Z},k,c} > \alpha, 0 \leq k \leq n\}$, satisfies

$$\Pr\left(n(c) \in \{n-k : p_{\boldsymbol{Z},k,c} > \alpha, 0 \leq k \leq n\}\right) = \Pr\left(p_{\boldsymbol{Z},n-n(c),c} > \alpha\right) = 1 - \Pr\left(p_{\boldsymbol{Z},n-n(c),c} \leq \alpha\right) \\ \geq 1 - \alpha,$$

where the last inequality holds because, from Theorem 3 and (15), $p_{\boldsymbol{Z},n-n(c),c}$ is a valid p-value for testing the null hypothesis $H_{n-n(c),c}$ of $n(c) \leq n - \{n - n(c)\}$ that always holds by definition. Moreover, because $p_{\boldsymbol{Z},k,c}$ is decreasing in k, the confidence set $\{n-k : p_{\boldsymbol{Z},k,c} > \alpha, 0 \leq k \leq n\}$ must have the form of $\{j : n - \overline{k} \leq j \leq n\}$ with $\overline{k} = \sup\{k : p_{\boldsymbol{Z},k,c} > \alpha, 0 \leq k \leq n\}$. Therefore, (c) in Corollary 2 holds. From the above, Corollary 2 holds. □

Proof of Theorem 5. First, we prove (17). We first show that if the left hand size of (17) holds, then the right hand side must also hold. For any $1 \leq k \leq n$, suppose $\tau_{(k)}$ is in $\{c : p_{\boldsymbol{Z},k,c} > \alpha, c \in \mathbb{R}\}$. This implies that $p_{\boldsymbol{Z},k,\tau_{(k)}} > \alpha$. For any $c \in \mathbb{R}$ with $p_{\boldsymbol{Z},k,c} \leq \alpha$, because the p-value $p_{\boldsymbol{Z},k,c}$ is increasing in c from Theorem (4), we must have $\tau_{(k)} > c$, or equivalently $\boldsymbol{\tau} \in \mathcal{H}_{k,c}^{\complement}$. Thus, $\boldsymbol{\tau} \in \bigcap_{c: p_{\boldsymbol{Z},k,c} \leq \alpha} \mathcal{H}_{k,c}^{\complement}$. We then show that if the left hand size of (17) fails, then the right hand side must also fail. For any $1 \leq k \leq n$, suppose $\tau_{(k)}$ is not in $\{c : p_{\boldsymbol{Z},k,c} > \alpha, c \in \mathbb{R}\}$. This implies that $p_{\boldsymbol{Z},k,\tau_{(k)}} \leq \alpha$. Because $\boldsymbol{\tau} \in \mathcal{H}_{k,\tau_{(k)}}$ by definition, we have $\boldsymbol{\tau} \in \mathcal{H}_{k,\tau_{(k)}} \subset \bigcup_{c: p_{\boldsymbol{Z},k,c} \leq \alpha} \mathcal{H}_{k,c}$, or equivalently $\boldsymbol{\tau} \notin \bigcap_{c: p_{\boldsymbol{Z},k,c} \leq \alpha} \mathcal{H}_{k,c}^{\complement}$. Therefore, (17) holds.

Second, we prove (18). We first show that if the left hand side of (18) holds, then the right hand side must also hold. For any $c \in \mathbb{R}$, suppose $n(c)$ is in $\{n-k : p_{\boldsymbol{Z},k,c} > \alpha, 0 \leq k \leq n\}$. This implies that $p_{\boldsymbol{Z},n-n(c),c} > \alpha$. For any k with $p_{\boldsymbol{Z},k,c} \leq \alpha$, because $p_{\boldsymbol{Z},k,c}$ is decreasing in k from Corollary 2, we must have $n - n(c) < k$, or equivalently $\boldsymbol{\tau} \in \mathcal{H}_{k,c}^{\complement}$. Thus, $\boldsymbol{\tau} \in \bigcap_{k: p_{\boldsymbol{Z},k,c} \leq \alpha} \mathcal{H}_{k,c}^{\complement}$. We then show that if the left hand side of (18) fails, then the right hand side must also fail. For any $c \in \mathbb{R}$, suppose $n(c)$ is not in $\{n-k : p_{\boldsymbol{Z},k,c} > \alpha, 0 \leq k \leq n\}$. This implies that $p_{\boldsymbol{Z},n-n(c),c} \leq \alpha$. Because $\boldsymbol{\tau} \in \mathcal{H}_{n-n(c),c}$ by definition, we have $\boldsymbol{\tau} \in \mathcal{H}_{n-n(c),c} \subset \bigcup_{k: p_{\boldsymbol{Z},k,c} \leq \alpha} \mathcal{H}_{k,c}$, or equivalently $\boldsymbol{\tau} \notin \bigcap_{k: p_{\boldsymbol{Z},k,c} \leq \alpha} \mathcal{H}_{k,c}^{\complement}$. Therefore, (18) holds.

Third, we prove that $\bigcap_{k,c:p_{\boldsymbol{Z},k,c}\le\alpha}\mathcal{H}_{k,c}^{\complement}$ is a $1-\alpha$ confidence set for the true treatment effect $\boldsymbol{\tau}$. The coverage probability of the set $\bigcap_{k,c:p_{\boldsymbol{Z},k,c}\le\alpha}\mathcal{H}_{k,c}^{\complement}$ has the following equivalent forms:

$$\Pr\Big(\boldsymbol{\tau}\in\bigcap_{k,c:p_{\boldsymbol{Z},k,c}\le\alpha}\mathcal{H}_{k,c}^{\complement}\Big)=1-\Pr\Big(\boldsymbol{\tau}\in\bigcup_{k,c:p_{\boldsymbol{Z},k,c}\le\alpha}\mathcal{H}_{k,c}\Big). \tag{A19}$$

If $\boldsymbol{\tau}\in\bigcup_{k,c:p_{\boldsymbol{Z},k,c}\le\alpha}\mathcal{H}_{k,c}$, then there must exists (k,c) such that $p_{\boldsymbol{Z},k,c}\le\alpha$ and $\boldsymbol{\tau}\in\mathcal{H}_{k,c}$. By definition, this implies that $p_{\boldsymbol{Z},\boldsymbol{\tau}}\le\sup_{\boldsymbol{\delta}\in\mathcal{H}_{k,c}}p_{\boldsymbol{Z},\boldsymbol{\delta}}\equiv p_{\boldsymbol{Z},k,c}\le\alpha$. Thus, from (A19), we have

$$\Pr\Big(\boldsymbol{\tau}\in\bigcap_{k,c:p_{\boldsymbol{Z},k,c}\le\alpha}\mathcal{H}_{k,c}^{\complement}\Big)=1-\Pr\Big(\boldsymbol{\tau}\in\bigcup_{k,c:p_{\boldsymbol{Z},k,c}\le\alpha}\mathcal{H}_{k,c}\Big)\ge1-\Pr\left(p_{\boldsymbol{Z},\boldsymbol{\tau}}\le\alpha\right).$$

By the validity of usual randomization test, $\Pr(p_{\boldsymbol{Z},\boldsymbol{\tau}}\le\alpha)\le\alpha$, and therefore,

$$\Pr\Big(\boldsymbol{\tau}\in\bigcap_{k,c:p_{\boldsymbol{Z},k,c}\le\alpha}\mathcal{H}_{k,c}^{\complement}\Big)\ge1-\Pr\left(p_{\boldsymbol{Z},\boldsymbol{\tau}}\le\alpha\right)\ge1-\alpha.$$

From the above, Theorem 5 holds. □

A6. Further simulation results

A6.1. Simulation demonstrating power of different Stephenson rank statistics

In the main paper's simulations we can see how power changes as a function of s. Extending those results, Figure A1, built from those same simulation results, shows the power of various Stephenson rank statistics for testing whether each quantile of individual effect is bounded by zero, i.e., $H_{k,0}:\tau_{(k)}\le0$ over all $1\le k\le n$, under the data generating model with $n=120$ and (τ_0,ω,ρ) equal to $(1,1,-0.9)$. (We omit those values of k for which all tests under consideration have zero power.) From Figure A1, we can see that $s=2$ is preferred for larger quantiles of individual effects, while $s=4$ is preferred for smaller quantiles.

A6.2. Simulation for heavy-tailed outcomes and individual effects

We next conduct a simulation with heavy-tailed outcomes and individual treatment effects. We will demonstrate that, compared to inference on average treatment effects (e.g., Neyman 1923), the proposed inference on quantiles does not require any large-sample approximation and can be more robust to outliers. In particular, we consider the case of generally positive constant individual effects with a few extreme negative outliers. In the next subsection, we also consider a scenario with heavy-tailed distributions of the potential outcomes under Fisher's null of no effect.

We simulate an experiment with 120 units, among which two thirds will be randomly assigned to treatment and the remaining to control. We assume the treatment increases a certain

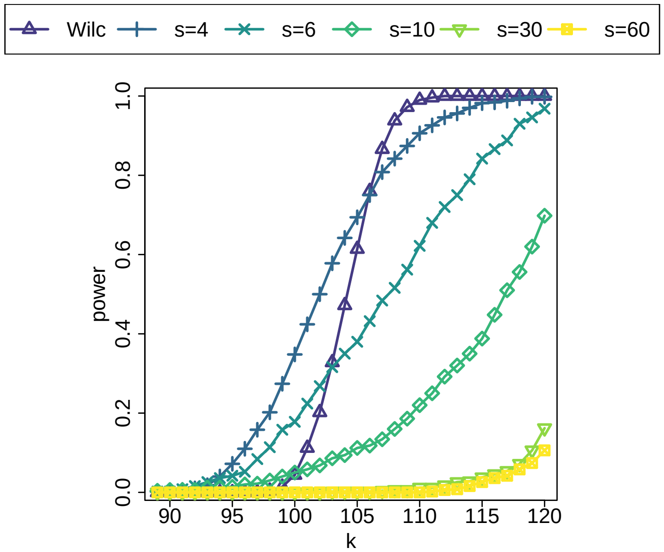


Figure A1: Power of various Stephenson rank statistics for testing the null hypothesis of $H_{k,0}$: $\tau_{(k)} \leq 0$ over all k, under model (20) with $n = 120$ and (τ_0, ω, ρ) equal to $(1, 1, -0.9)$.

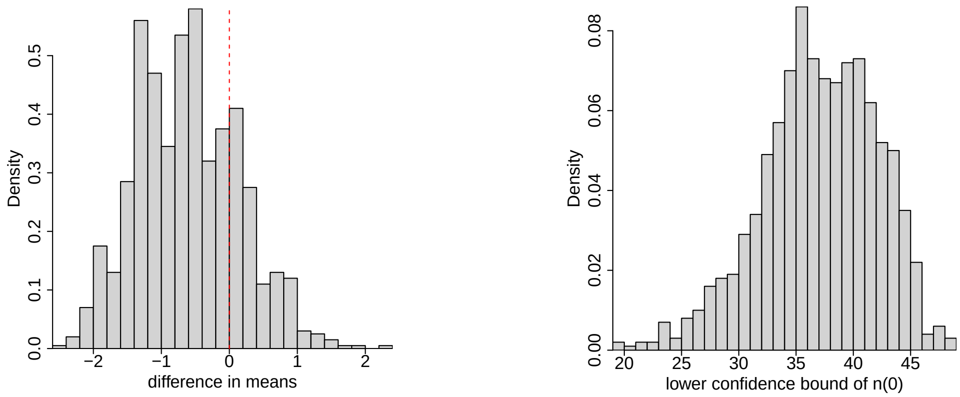
risk factor by 2 for 95% of the units, and decreases it by 50 for the remaining 5% of units. We simulate the control potential outcomes from the standard normal distribution. Once generated, all the potential outcomes are fixed. For the final sample, about 5% of the units receive large benefit, but the remainder would incur a harmful effect. The true average treatment effect is negative, indicating an average benefit, even though the majority of units are harmed.

Figure A2(a) shows the histogram of the sampling distribution of the usual difference-in-means estimator. Over all the simulated assignments, the difference-in-means estimator is negative about 77% of the time, and its average is close to the true average effect -0.6. In general, the difference in means estimator would indicate that the treatment is reducing overall risk level and is apparently beneficial. However, such a conclusion would potentially be misleading, because the treatment is harmful for most units.

Figure A2(b) shows the histogram of the 90% lower confidence limit of the number of units with higher risk level under treatment than control (i.e., $n(0)$), based on the Stephenson rank statistic with $s = 6$. Over all simulated assignments, the lower confidence limit of $n(0)$ has an average value of 37.5 (about 31% of the units), and can sometimes reach 49 (about 41% of the units). Our method reliably detects that the treatment harms a significant amount of units; such a finding would signal to researchers that the treatment would need further careful investigation despite an apparent average benefit.

A6.3. Simulation for heavy-tailed outcomes under Fisher's null

We simulate the potential outcomes $Y_i(1) = Y_i(0)$'s from a skewed t distribution with degrees of freedom 1.5 and skewing parameter 5 (see, e.g., Azzalini 2020), under which all the individual treatment effects are zero, i.e., Fisher's null H_0 holds. To mimic the finite population inference,



(a) (b)

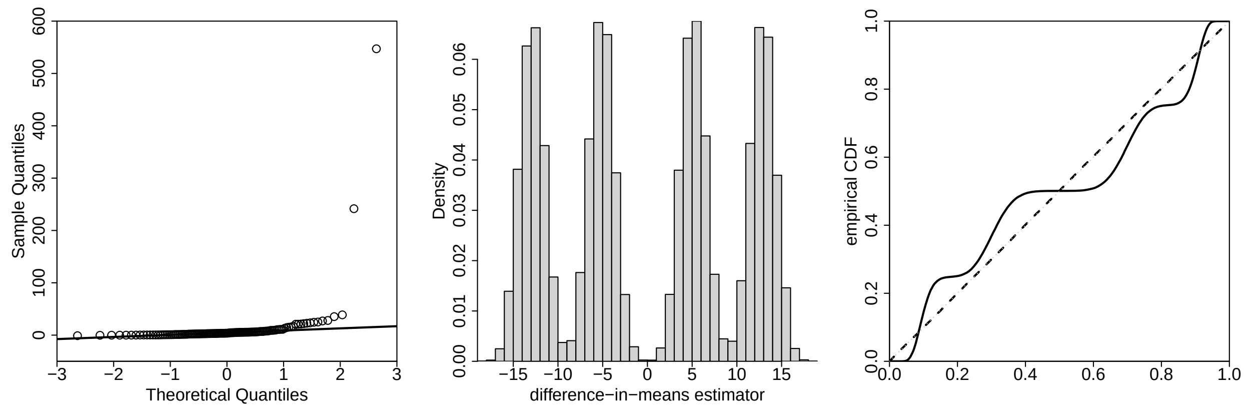
Figure A2: Simulation when individual treatment effects have extreme values or outliers. (a) shows the histogram of the usual difference-in-means estimator. (b) shows the histogram of the 90% lower confidence limit of the number of units with positive effects using Theorem 4.

all the potential outcomes are kept fixed once generated. We conduct simulation under a CRE with $n = 120$ units, where half of the units are assigned to each treatment group. Figure A3(a) shows the normal Q-Q plot of the control potential outcomes, from which we can see that the potential outcomes are right-skewed and heavy-tailed. Figure A3(b) shows the histogram of the difference-in-means estimator under the CRE, which has multiple modes. This implies that the large-sample normal approximation as in Neyman (1923) may work poorly with heavy-tailed outcomes. Consequently, it is not surprising to see that the p-value using the two-sample t-statistic and normal approximation does not control the type-I error well, as shown in Figure A3(c).[2] From the same figure, the Fisher randomization p-value using the Stephenson rank sum statistic with $s = 10$ is almost uniformly distributed on $(0, 1)$. Moreover, this p-value is also valid for testing the bounded null $H_{\preccurlyeq \mathbf{0}}$.

A6.4. Simulation with varying sample size

We conduct the same simulation as in Section 7.2, except that we vary the sample size n from 240 to 960. Figures A4–A6 show the 90% lower confidence limit for all quantiles of individual effects, averaging over all simulated treatment assignment, when the sample size n equals 240, 480 and 960, respectively. The implications from these simulation results are mostly the same as that from Section 7.2. First, when the average effect is non-positive, the Wilcoxon rank has almost no power to detect any positive individual effects, while the Stephenson rank with larger s tends to give more informative results. Second, when the average treatment effect is positive, too large an s for Stephenson rank can deteriorate the power, especially when the individual treatment effect

[2] Wu and Ding (2018) and Cohen and Fogarty (2020) suggest using the permutation distribution of the t-statistic as the reference distribution for calculating the p-value. However, since the t-statistic is in general neither effect increasing nor differential increasing, Theorem 1(a) cannot guarantee the resulting p-value to be valid for testing the bounded null.



(a) (b) (c)

Figure A3: Simulation under skewed and heavy-tailed outcomes. (a) shows the normal Q-Q plot of the potential outcomes $Y_i(1) = Y_i(0)$'s. (b) shows the histogram of the difference-in-means estimator under a CRE. (c) shows the empirical distribution functions of the p-value from Neyman (1923) using t-statistic with normal approximation (denoted by the solid line) and that from Fisher randomization test using Stephenson rank statistic with $s = 10$ (denoted by the dashed line).

is negatively correlated with the control potential outcome. It will be interesting to theoretically investigate the role of s for inferring quantiles of individual effects, and we leave it for future study.

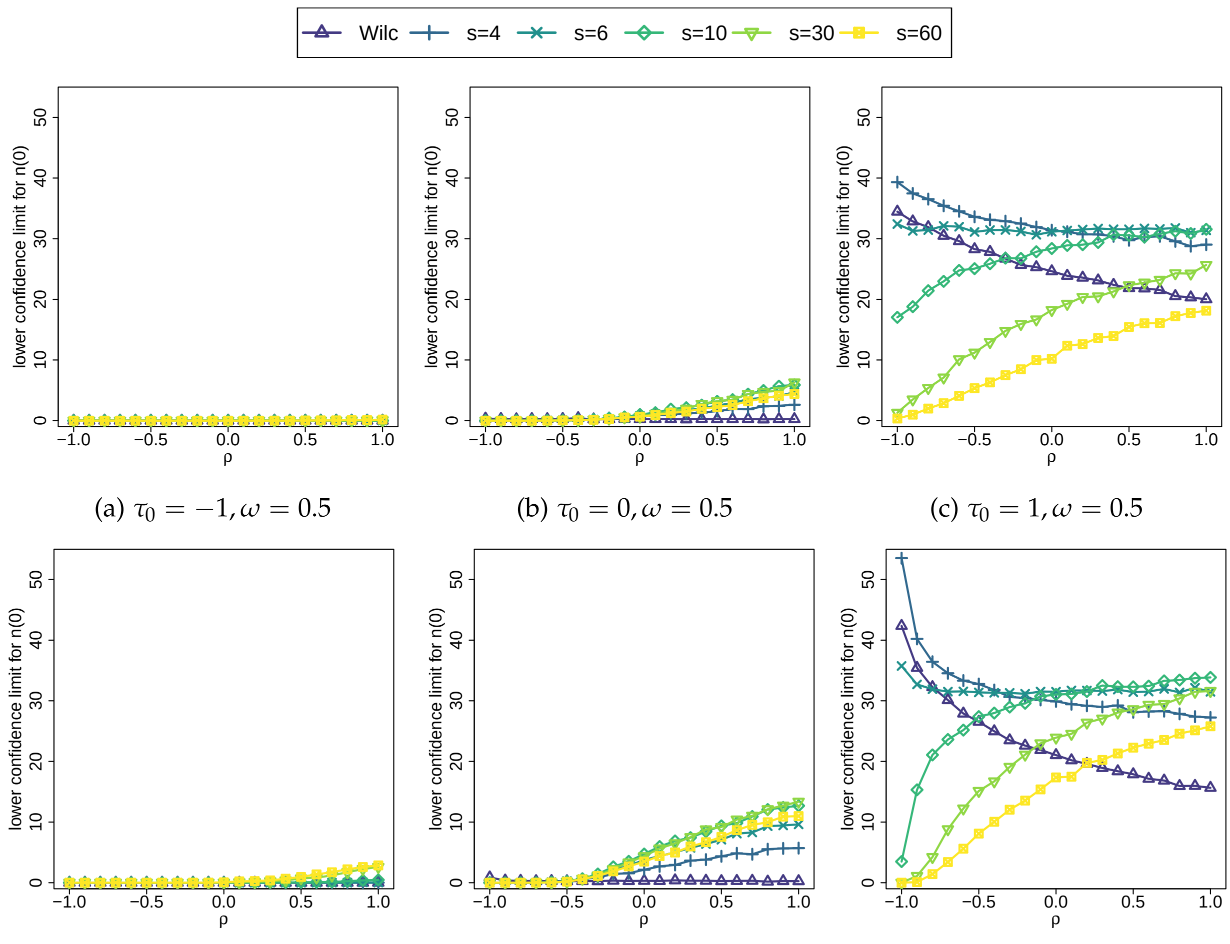
A7. Further Applications

A7.1. Selecting s for the Professional Development Application

The main paper outlines a series of power simulations across a family of data generating processes to see how well different s values for the Stephenson rank test performed for detecting different numbers and types of treatment effects. We next look into these results in more detail.

To run these simulations we use a method in the `RIQITE` package, `explore_stephenson_s`, that runs the simulations for us. We provide the empirical distribution of the control-side outcomes as the reference distribution, and specify a variety of treatment impact models that are arguably consistent with the difference in final distributions between treatment and control. For example, we specify impact models with a true average treatment impact in line with our estimate from the data.

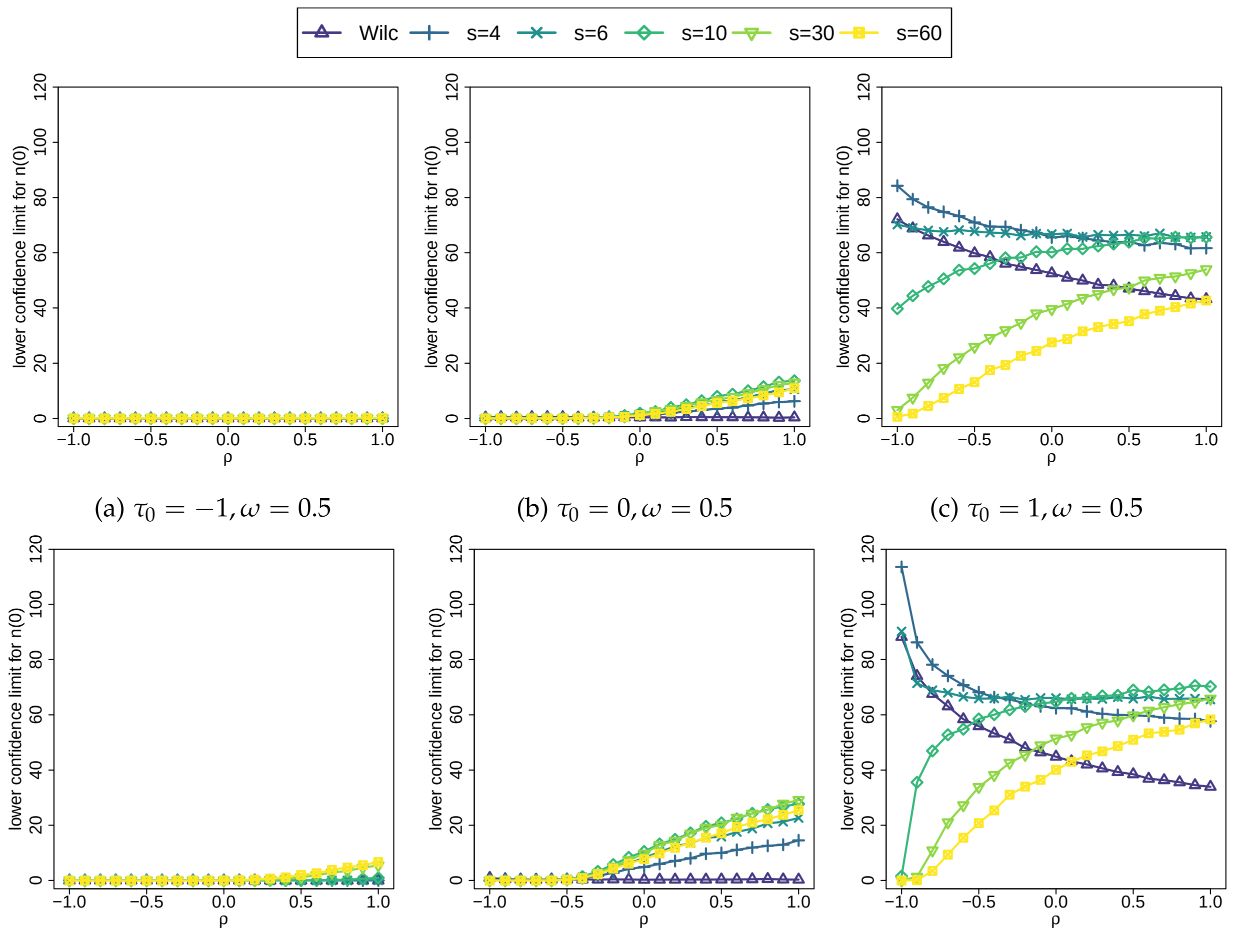
Figure A7 shows median confidence bounds over different quantiles as a function of s. We grouped the top 100 quantiles into groups of top 10, next 20, and next 30, and plot the median of the median bound across the quantiles in each group (the last 40 are not shown as the intervals were generally not informative). Generally, the curves are flat, showing that there is some latitude for selecting s across the contexts explored. Overall, smaller s, such as $s = 6$, seem preferred.



(a) $\tau_0 = -1, \omega = 0.5$ (b) $\tau_0 = 0, \omega = 0.5$ (c) $\tau_0 = 1, \omega = 0.5$

(d) $\tau_0 = -1, \omega = 1$ (e) $\tau_0 = 0, \omega = 1$ (f) $\tau_0 = 1, \omega = 1$

Figure A4: Average lower limits of 90% confidence intervals for the number of units with positive effects $n(0)$. The potential outcomes are generated from (20) with sample size $n = 240$ and different values of (τ_0, ω, ρ).



(a) $\tau_0 = -1, \omega = 0.5$

(b) $\tau_0 = 0, \omega = 0.5$

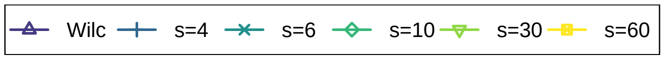
(c) $\tau_0 = 1, \omega = 0.5$

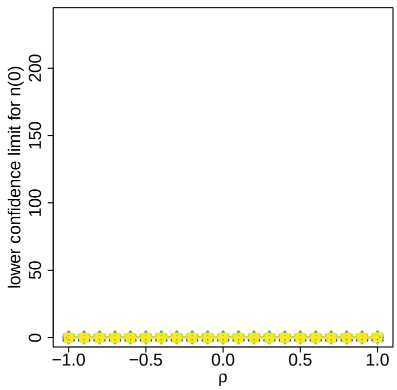
(d) $\tau_0 = -1, \omega = 1$

(e) $\tau_0 = 0, \omega = 1$

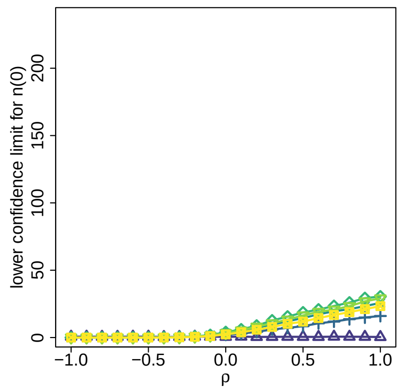
(f) $\tau_0 = 1, \omega = 1$

Figure A5: Average lower limits of 90% confidence intervals for the number of units with positive effects $n(0)$. The potential outcomes are generated from (20) with sample size $n = 480$ and different values of (τ_0, ω, ρ).

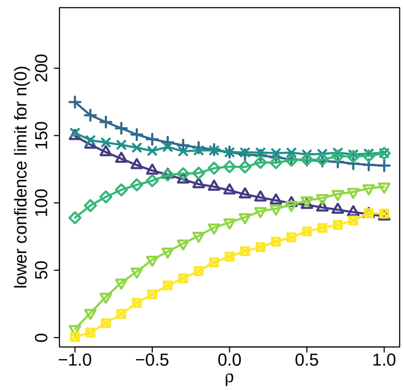




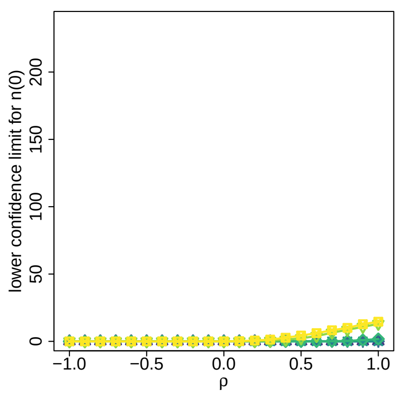
(a) $\tau_0 = -1, \omega = 0.5$



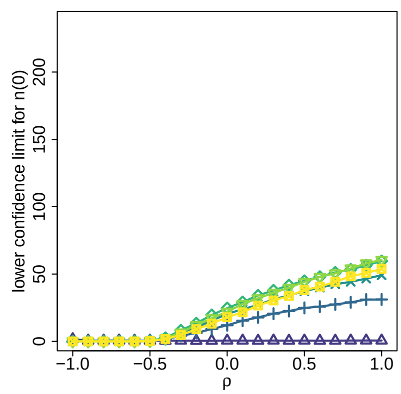
(b) $\tau_0 = 0, \omega = 0.5$



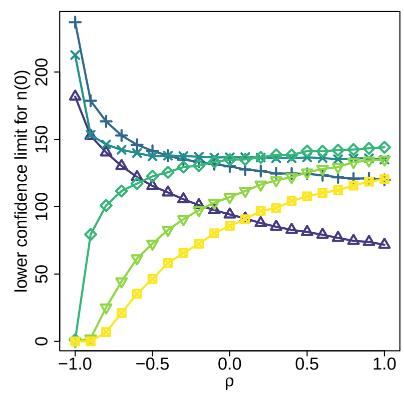
(c) $\tau_0 = 1, \omega = 0.5$



(d) $\tau_0 = -1, \omega = 1$



(e) $\tau_0 = 0, \omega = 1$



(f) $\tau_0 = 1, \omega = 1$

Figure A6: Average lower limits of 90% confidence intervals for the number of units with positive effects $n(0)$. The potential outcomes are generated from (20) with sample size $n = 960$ and different values of (τ_0, ω, ρ).

See the code file for more specifics on how these results are generated; that file is designed to provide code to make conducting similar simulations on other datasets very straightforward.

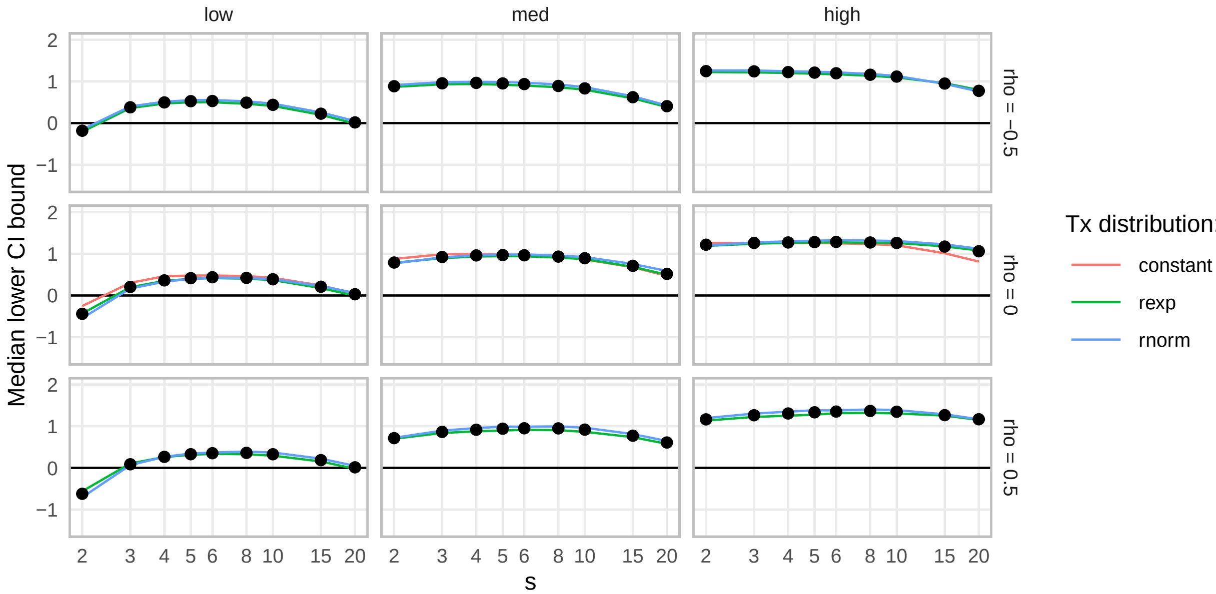


Figure A7: Median CI lower bounds over different groups of quantiles and simulation scenarios as a function of s for an exploratory simulation calibrated to the teacher professional development data. From right to left we have top 10 quantiles, next 20, and next 30. Rows correspond to different correlations of $Y_i(0)$ and τ_i. Black dots give empirical averages across all scenarios considered.

We can also look at the number of units found to be significant across the different scenarios and see which s values identify the most units as significant, on average. Figure A8 shows how $s = 8$ seems to maximize, although $s = 6$ or $s = 10$ perform quite similarly.

We are facing somewhat of a trade-off: more units being tagged as significant calls for higher s, and more informative bounds on the higher end of the distribution calls for slightly lower s.

A7.2. Testing monotonicity of an instrumental variable

The assumption that the instrument has monotonic effects on the treatment, though conventionally invoked for identification of instrumental variable (IV) estimates (Angrist et al. 1996), is rarely evaluated in empirical applications. Recently, however, the issue of non-monotonicity has received attention in the active literature on school-entry age, in which numerous studies involve instruments based on laws regulating entry age (Aliprantis 2012; Barua and Lang 2016; for an overview, see Fiorini and Stevens 2014). Typically, these laws select an arbitrary date of birth before which children are allowed to enter school in a given calendar year. If the cutoff date is January 1, for example, most children born in December will be about 11 months younger when they enter school than children born in January. Due to imperfect compliance with the instrument, however, some fraction of December children may "redshirt" and start the following

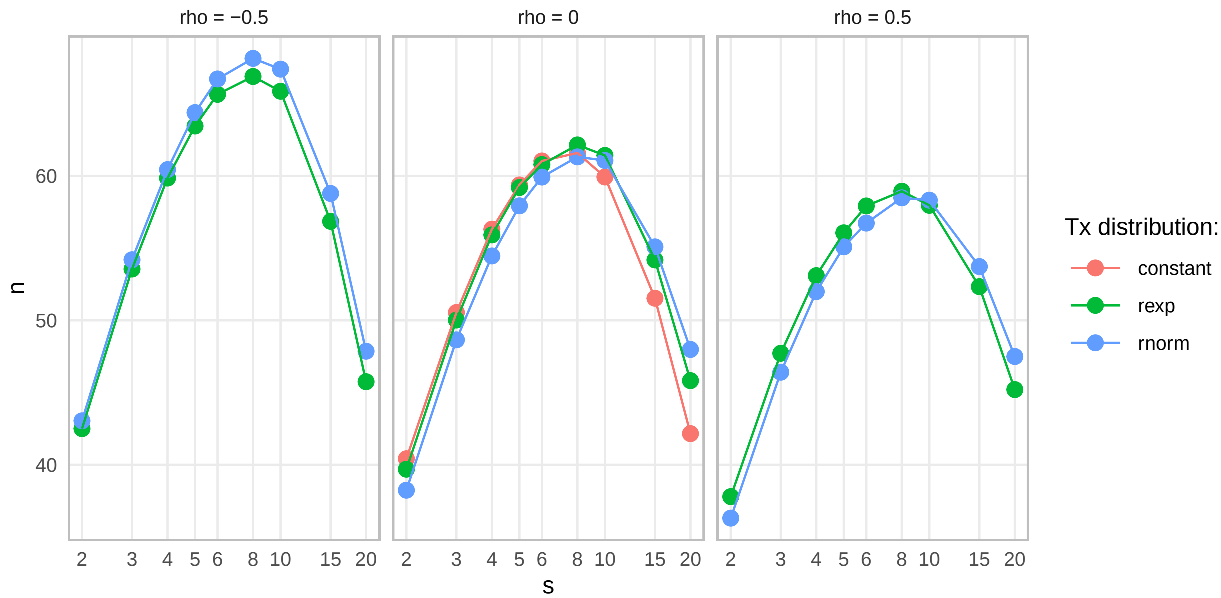


Figure A8: Average number of units found to be significant as a function of s.

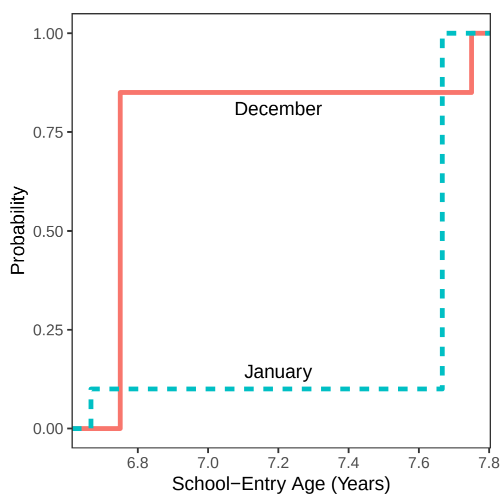


Figure A9: Empirical cumulative distribution functions of school-entry age for units born in December and January, respectively (Black et al. 2011).

school year, at which time they will be one month *older* than January children who started on time. Unless the December children who redshirt would also have redshirted had they been born in January, monotonicity is violated. That is, the effect of December birth on school-entry age is typically negative, but for a few children it is positive (an analogous logic holds for January children who start early).

For a simple illustration of how randomization inference can be used to evaluate monotonicity, we re-analyze data on 104,000 children born in December or January from Black et al. (2011)'s IV study of school-entry age, which have previously been analyzed from a sampling-based perspective by Fiorini and Stevens (2014)[3]. The latter authors note that the cumulative distribution functions of December- and January-born children in these data cross each other, suggesting a violation of monotonicity. Figure A9 indicates this clearly. Most children started school at an older age if they were born in January rather than December. Indeed, this first-stage relationship is incredibly strong, with an average effect of -0.667 years and an F statistic of 106,256. This would conventionally be considered persuasive evidence of a valid instrument. Note, however, that at the tails of the distribution the relationship between month of birth and entry age reverses: January birthdays predominate among the youngest starters, and December does so among the oldest.

Table A1: Randomization tests for the bounded null $H_{\preccurlyeq \mathbf{0}}$, which is equivalent to the monotonicity assumption. The intervals are 90% confidence intervals for the maximum individual effect, constructed by inverting the corresponding randomization tests. Columns 2–4 show results from randomization inference using different test statistics. The last column shows results from the classical Student's t-test.

	Difference-in-means	Wilcoxon	Stephenson	Student's t-test
p-value	1	1	0	1
90% CI	$[-0.669, \infty)$	$[-0.916, \infty)$	$[0.084, \infty)$	$[-0.670, \infty)$

If we designate December birth as the assigned-to-treatment condition and January birth as control, then the monotonicity assumption is equivalent to the null hypothesis $H_{\preccurlyeq \mathbf{0}}$: being born in December did not cause any child to go to school at an older age than they would have if born in January. We assume that birth month is as-if randomly assigned, and conduct randomization inference for the effect of December birth on the school-entry age. We test the bounded null $H_{\preccurlyeq \mathbf{0}}$ using the randomization p-value $p_{\boldsymbol{Z},\mathbf{0}}$ with various test statistics satisfying the conditions in Theorem 1(a), including the difference-in-means, Wilcoxon rank sum and Stephenson rank sum with $s = 10$. Table A1 lists the results from randomization tests using these three test statistics, supplemented by the classical Student's t-test. We emphasize that, although both the randomization p-values and the corresponding intervals are numerically the same as that under the usual

[3]Table 1 of Black et al. (2011) cross-tabulates month of birth and school entry age (early/on-time/late) in terms of proportions, and Table 3 there reports the total number of subjects born in January or December (the "discontinuity subsample"): 104,023. Similar to Fiorini and Stevens (2014), we assume equal numbers of subjects born in January and December, and round the total number of subjects to 104,000, to ensure integer number of subjects in each subgroup.

constant treatment effect assumption, they are also valid p-values for the bounded null $H_{\preccurlyeq \mathbf{0}}$ and valid confidence intervals for the maximum individual effect $\tau_{\max}$, as demonstrated in Section 4. Because the difference-in-means estimator for the average effect is negative, from the simulation results in Section 7.1, it is not surprising that neither difference-in-means or Wilcoxon rank sum give significant p-values. However, the Stephenson rank sum gives an almost zero p-value, strong evidence of the existence of units violating the monotonicity assumption. Intuitively, the significant p-value is driven by the Stephenson rank placing more weight on larger outcomes coupled with 15% of December-born children having a school-entry age of 7.75, which is larger than the maximum school-entry age 7.67 for the January-born children. This means the treatment group has a large share of the most extreme observations. The corresponding 90% lower confidence limit is 0.084 year, or equivalently about 1 month, suggesting some children would first enter the school one-month older if born in December than in January. We can therefore confidently conclude that being born in December increased school-entry age for at least some students, i.e., the IV monotonicity assumption is violated in this application.

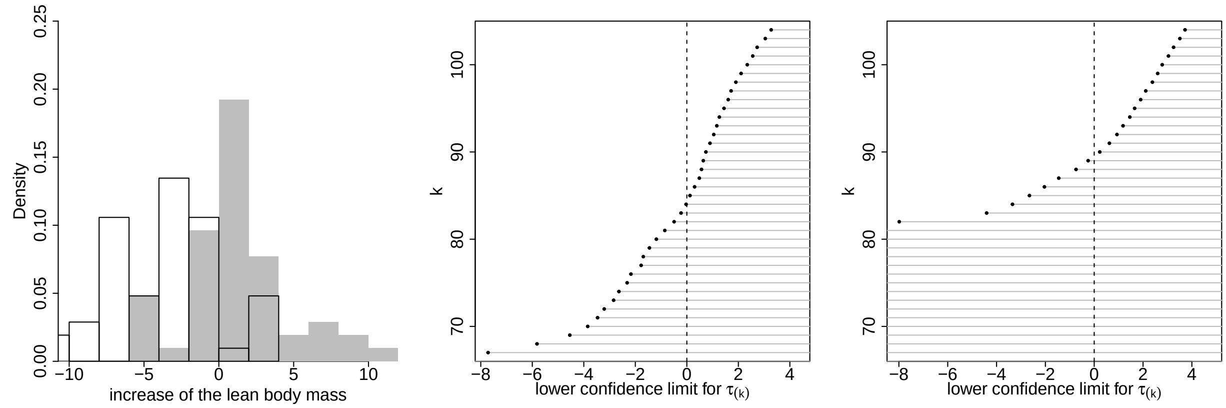
We now apply Theorem 6 to study the effect range. Using the Stephenson rank sum statistic with $s = 10$, the 95% upper confidence limit for the minimum effect of December birth is -0.917 years (-11 months) while the 95% lower confidence limit for the maximum effect is 0.084 years (1 month). We are therefore 90% confident that the range of the effect of birth month on school-entry age is at least 1 year, indicating significant individual effect heterogeneity. This of course is hardly surprising given Figure A9, which shows that despite the negative average effect of December birth, the children with the oldest school-entry age were born in this month.

A7.3. Evaluating the effects of six-month nutrition therapy

The homefood study (Blondal 2021) is a recent randomized controlled clinical trial that tries to evaluate the effects of home delivered food and nutrition therapy for discharged geriatric hospital patients.[4] Participants were randomized into two groups, and those in the treated group will be given free food for 24 weeks to fulfill protein and energy needs, which are based on individualized nutrition care plans designed by the dietitians. Here we focus on the effect of the treatment on the increase of lean body mass (kg) measured before and after the trial. We exclude two units with missing outcomes, resulting in 52 treated units and 52 control units. Figure A10(a) shows the histograms of the lean body mass changes in treated and control groups, respectively.

We then infer the lower confidence limits for all quantiles of individual treatment effects. Figures A10(b) and (c) show the 90% lower confidence limits for all quantiles of individual effects using Stephenson rank statistics with $s = 6$ and $s = 2$ (i.e., Wilcoxon rank), respectively. Obviously, the Stephenson rank with $s = 6$ gives more informative results, based on which we are 90% confident that at least 19.2% units would have higher lean body mass if receiving treatment instead of control.

[4]The details of the study can be found at `https://clinicaltrials.gov/ct2/show/NCT03995303`, and the data are publicly available at Harvard Dataverse with link `https://doi.org/10.7910/DVN/38X3LX`.



(a) (b) $s = 6$ (c) $s = 2$

Figure A10: Histograms of observed lean body mass increase and 90% confidence intervals for quantiles of individual effects. (a) shows the histograms of observed lean body mass increase in treatment (grey) and control (white) groups, respectively. (b) and (c) shows the 90% lower confidence limits for all quantiles of individual effects using the Stephenson rank statistics with s equals 6 and 2, respectively. The uninformative confidence intervals of $(-\infty, \infty)$ for quantiles of lower ranks are omitted from (b) and (c).

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